
Quantum-Enhanced Early Health Detection

From Current Research Gaps to Our Proposed Prototype

A single, printable study + research-gap + prototype-design + implementation + SIH-defence document, synthesised from ten attached research papers.

How to read this document. Parts 1–6 build the picture: field overview, state of the art, gap analysis, candidate novelties, our chosen research idea, our architecture. Parts 7–19 are the teaching core (classical ML → quantum basics → encoding → feature maps → QSVM → QNN/VQC → data re-uploading → barren plateaus → quantum advantage → healthcare metrics → robustness → explainability → quantum optimisation). Parts 20–26 are the experimental plan, MVP/advanced split, roadmap, tech stack, code concepts and the written research contribution. Parts 27–30 prepare the SIH pitch, the judge Q&A, the paper map and the cheat sheets.

The ten source papers (referred to by key throughout)

Every major claim in this document is tagged with the paper key(s) that support it. Where an idea comes from outside this set, it is placed in a **RECENT LITERATURE** / **EXTERNAL EVIDENCE** box with a citation.

Key	Paper
[P1]	J.R. McClean, S. Boixo, V.N. Smelyanskiy, R. Babbush, H. Neven, “Barren plateaus in quantum neural network training landscapes,” <i>Nature Communications</i> 9 , 4812 (2018). arXiv:1803.11173.
[P2]	A. Pérez-Salinas, A. Cervera-Liarta, E. Gil-Fuster, J.I. Latorre, “Data re-uploading for a universal quantum classifier,” <i>Quantum</i> 4 , 226 (2020). arXiv:1907.02085.
[P3]	H.-Y. Huang, M. Broughton, M. Mohseni, R. Babbush, S. Boixo, H. Neven, J.R. McClean, “Power of data in quantum machine learning,” <i>Nature Communications</i> 12 , 2631 (2021). arXiv:2011.01938.
[P4]	L.M. Donaire, G. Ortega, F. Orts, E. Martín Garzón, E. Filatovas, “A hybrid quantum-classical approach for liver disease detection using quantum machine learning,” <i>Engineering Applications of Artificial Intelligence</i> 164 (2026) 113240. DOI 10.1016/j.engappai.2025.113240. Code: github.com/LauraMDonaire/QML-Liver.
[P5]	A.E. Setiawan <i>et al.</i> , “A systematic literature review of quantum machine learning for medical: trends, datasets, topics, and methods,” <i>International Journal of Cognitive Computing in Engineering</i> 7 (2026) 609–630. DOI 10.1016/j.ijcce.2026.05.002. (94 primary studies, 2020–Apr 2025.)
[P6]	F.M. Creevey, J.A. Heredge, M.E. Sevier, L.C.L. Hollenberg, “Kernel Alignment for Quantum Support Vector Machines Using Genetic Algorithms” (the GEKO method), arXiv:2312.01562v2 (2026).
[P7]	G. Zorlu, C. Colak, “A Robustness-Oriented Quantum–Classical Hybrid Machine Learning Pipeline for Breast Cancer Diagnosis: External Validation, Explainability, and Rigorous Benchmarking in the NISQ Era,” <i>Diagnostics</i> 16 (2026) 1996. DOI 10.3390/diagnostics16131996.
[P8]	A. Al Khan, E. Usupova, “Quantum Machine Learning Algorithms for Genome Disease Diagnosis,” Authorea preprint, 30 Sep 2025. DOI 10.22541/au.175924949.93262371/v1. <i>Not peer-reviewed; “Data may be preliminary.” Treated here as a low-rigour example.</i>
[P9]	A. Zeguendry, Z. Jarir, M. Quafafou, “Quantum Machine Learning: A Review and Case Studies,” <i>Entropy</i> 25 (2023) 287. DOI 10.3390/e25020287.
[P10]	R.S. Gupta, C.E. Wood, T. Engstrom, J.D. Pole, S. Shrapnel, “A systematic review of quantum machine learning for digital health,” <i>npj Digital Medicine</i> 8 (2025) 237. DOI 10.1038/s41746-025-01597-z. (4915 studies screened, 16 synthesised.)

ONE-LINE SUMMARY OF THE EVIDENCE BASE

Two independent 2025–2026 systematic reviews [P5][P10] and one rigorous 2026 benchmarking study [P7] all reach the same verdict: *after fair comparison against well-tuned classical models, there is no consistent, statistically significant quantum-machine-learning advantage on tabular / small-sample health data today.* The value that is realistically available

to a student team is **methodological**: a leakage-free, externally-validated, explainable, noise-tested pipeline in which a quantum model is compared honestly with a strong classical baseline — plus one narrow, defensible technical contribution on top.

Contents

The ten source papers	1
1 Field overview — the plain-language picture	8
1.1 Classical machine learning in healthcare	8
1.2 Quantum computing (only what we need)	8
1.3 Quantum machine learning (QML)	8
1.4 Hybrid quantum–classical ML	8
1.5 QSVM, QNN/VQC, quantum kernels, quantum feature maps, quantum optimisation — one paragraph each	9
1.6 Why researchers are interested	9
1.7 Where is the field actually standing in 2026?	9
2 Current state of the art — a literature map	10
2.1 1. Classical ML + quantum preprocessing / quantum feature engineering	10
2.2 2. QSVM / quantum kernel methods	10
2.3 3. Variational Quantum Classifiers (VQC)	10
2.4 4. Quantum Neural Networks (QNN), incl. data re-uploading	11
2.5 5. Hybrid (classical deep net + quantum layer)	11
2.6 6. Quantum feature maps (as an object of study)	11
2.7 7. Quantum kernel / feature-map optimisation	11
2.8 8. Quantum optimisation methods (annealing / QUBO)	11
2.9 9. Quantum-enhanced medical imaging	12
2.10 10. Quantum-enhanced tabular health prediction	12
2.11 11. Quantum-enhanced biosignal analysis	12
2.12 12. Explainable quantum ML	12
2.13 13. Noise-aware / hardware-aware QML	12
2.14 14. Robustness and generalization	12
3 Literature gap analysis	13
3.1 The gap table	13
3.2 Which gaps are already being addressed by recent papers?	15
4 What should our novelty be? — candidate directions	16
4.1 Direction A — Leakage-free, externally-validated evaluation of an <i>optimised</i> quantum feature map / kernel for early health detection	16
4.2 Direction B — Is a quantum model genuinely more robust to class imbalance? A clean test	17
4.3 Direction C — Uncertainty-aware quantum early-detection with a calibrated risk output	17
4.4 Direction D — Noise-budget characterisation for a tabular early-detection circuit	17
4.5 Direction E — Quantum-enhanced early detection on a biosignal (ECG/EEG) with a classical feature extractor + optimised quantum head	18
4.6 Ranking	18
5 Our proposed research idea	18
5.1 The exact problem we are solving that existing systems do not	18
5.2 Research Problem	19
5.3 Research Gap	19
5.4 Hypothesis	19
5.5 Proposed Contribution	19

5.6	Expected Benefit	19
5.7	Limitations (state these before a judge does)	19
5.8	What would falsify our hypothesis?	20
6	Proposed architecture	20
6.1	Architecture diagram	20
6.2	Every block explained	21
7	Classical ML foundation (only what the prototype needs)	23
7.1	Data objects	23
7.2	Preparing data	23
7.3	Choosing and shrinking features	24
7.4	Evaluating honestly	24
7.5	The models we will actually use	25
7.6	Evaluation metrics	25
8	Quantum basics (from absolute zero, only what we need)	26
8.1	Bit vs. qubit	26
8.2	Superposition	26
8.3	Measurement, probability amplitudes, expectation values, shots	26
8.4	Quantum state & Bloch-sphere intuition	27
8.5	Quantum gates	27
8.6	Entanglement	27
8.7	Quantum circuits & parameterised (variational) circuits	27
8.8	Noise & NISQ	27
8.9	Quantum simulators vs. real hardware	28
9	Data encoding — classical numbers into a quantum state	28
9.1	Why classical data must be encoded	28
9.2	The four encodings we care about	28
9.3	Worked example: a five-feature patient	29
9.4	Dimensionality constraints & the PCA–qubit link	29
10	Quantum feature maps	29
10.1	Build-up: what is a feature map in ordinary ML?	30
10.2	What changes when the feature space is quantum?	30
10.3	Ingredients of a quantum feature map	30
10.4	Expressivity vs. trainability — the central trade-off	31
10.5	Feature-map selection	31
10.6	Hardware-aware feature maps	31
11	QSVM — the deepest section	31
11.1	Start from classical SVM (intuition before maths)	31
11.2	The pieces	31
11.3	Nonlinear classification & the kernel trick	32
11.4	From classical kernel to quantum kernel	32
11.5	Estimating one kernel entry	32
11.6	Training & prediction	32
11.7	The quantum / classical split	32
11.8	Limitations (say them plainly)	33
11.9	Complete tiny healthcare workflow	33
12	QNN / VQC — the trainable quantum circuit	33

12.1 One-sentence idea	33
12.2 The parts	33
12.3 The training loop	34
12.4 Prediction	34
12.5 Overfitting & trainability notes	34
13 Data re-uploading (from [P2])	35
13.1 What it is	35
13.2 Why it was proposed	35
13.3 Architecture	35
13.4 Repeated encoding & trainable layers	35
13.5 One-qubit intuition → multi-qubit	36
13.6 Advantages / limitations	36
14 Barren plateaus (from [P1])	36
14.1 The concepts	36
14.2 Contrast with classical deep nets	37
14.3 What this means for OUR QNN — practical design rules	37
15 Quantum advantage — and how we would actually test for it (from [P3])	37
15.1 The vocabulary, carefully	37
15.2 Why accuracy alone is not evidence of advantage	38
15.3 The role of data, representation, kernel choice, baseline	38
15.4 [P3]’s testing methodology (the tools we adopt)	38
15.5 How WE would test whether our quantum approach helps	39
16 Healthcare-specific issues	39
16.1 Why early detection is not ordinary classification	39
16.2 The metric set (with the clinical meaning)	39
16.3 Privacy, governance, clinical validation	40
17 Robustness — the experiments	40
17.1 Uncertainty estimation (block 10)	41
18 Explainability	41
18.1 Why it matters	41
18.2 Feature importance — global and local	41
18.3 Explaining a quantum model (the hard part)	42
18.4 Limitations of explaining quantum models	42
18.5 Our explainable dashboard output — designed responsibly	42
19 Quantum optimisation — GEKO and where it belongs	43
19.1 Why a fixed quantum kernel may not be optimal	43
19.2 Kernel alignment (the cheap optimiser — our MVP)	43
19.3 GEKO — genetic kernel optimisation (the advanced optimiser, [P6])	43
19.4 Should GEKO be in our MVP?	44
19.5 Kernel alignment $\neq$ accuracy (a caution)	44
20 Experiment design	44
20.1 The three-way comparison	44
20.2 Protocol (identical for every arm)	44
20.3 Experiment table	45

21 Ablation study	45
21.1 What each component is expected to contribute	46
22 MVP vs. advanced version	47
22.1 MVP — realistically buildable during SIH	47
22.2 Advanced version — research-oriented extensions	47
22.3 Feature / priority table	47
23 Implementation roadmap	48
23.1 Phase 0 — Literature & gap validation (0.5 week)	48
23.2 Phase 1 — Dataset & preprocessing (1 week)	48
23.3 Phase 2 — Classical baseline (1 week)	48
23.4 Phase 3 — Quantum environment (0.5 week)	48
23.5 Phase 4 — Quantum feature map (1 week)	48
23.6 Phase 5 — QSVM + map optimisation (1.5 weeks)	49
23.7 Phase 6 — QNN / VQC (1 week, can parallelise with Phase 5)	49
23.8 Phase 7 — Proposed novelty consolidation + GEKO (advanced) (1 week)	49
23.9 Phase 8 — Robustness (1 week)	49
23.10 Phase 9 — Explainability (0.5 week)	49
23.11 Phase 10 — Dashboard (1 week)	49
23.12 Phase 11 — Final experiments & write-up (1 week)	49
24 Tech stack	50
25 Code concepts (understand before you implement)	51
25.1 Programming concepts you must be solid on first	51
25.2 Small examples (skeletons, not production)	51
26 Research contribution (academic phrasing)	54
26.1 Problem Statement	54
26.2 Existing Limitation	54
26.3 Research Gap	54
26.4 Proposed Method	54
26.5 Novelty	54
26.6 Hypothesis	55
26.7 Experimental Validation	55
26.8 Expected Contribution	55
26.9 Limitations	55
27 SIH presentation — the research story	55
27.1 The narrative arc (12 beats)	55
27.2 2-minute explanation (say this)	56
27.3 5-minute explanation (structure + extra content)	56
28 Judge questions — 45 hard ones, answered honestly	57
28.1 “Why quantum at all?”	57
28.2 Hardware & scale	58
28.3 Design choices	58
28.4 Evaluation rigour	59
28.5 Scope, safety, novelty	60
29 Research paper → what I need from it	62
29.1 [P1] Barren plateaus in QNN training landscapes (McClean et al. 2018)	62

29.2	[P2] Data re-uploading for a universal quantum classifier (Pérez-Salinas et al. 2020)	63
29.3	[P3] Power of data in quantum machine learning (Huang et al. 2021)	63
29.4	[P4] Hybrid quantum-classical liver disease detection / QML-Liver (Donaire et al. 2026)	63
29.5	[P5] SLR of QML for medical (Setiawan et al. 2026)	63
29.6	[P6] Kernel Alignment for QSVM using Genetic Algorithms / GEKO (Creevey et al. 2026)	63
29.7	[P7] Robustness-Oriented Quantum–Classical Hybrid Pipeline for Breast Cancer (Zorlu & Colak 2026)	64
29.8	[P8] Quantum ML for Genome Disease Diagnosis (Al Khan & Usupova, preprint 2025)	64
29.9	[P9] Quantum Machine Learning: A Review and Case Studies (Zeguendry et al. 2023)	64
29.10	[P10] Systematic review of QML for digital health (Gupta et al. 2025, npj Digital Medicine)	64
30	Final cheat sheets	65
	Cheat sheet 1 — Quantum ML terminology	65
	Cheat sheet 2 — QSVM	65
	Cheat sheet 3 — QNN/VQC	66
	Cheat sheet 4 — Quantum feature maps	66
	Cheat sheet 5 — Quantum kernels	66
	Cheat sheet 6 — Healthcare metrics	66
	Cheat sheet 7 — Research gaps	67
	Cheat sheet 8 — Our architecture	67
	Cheat sheet 9 — Roadmap	67
	Cheat sheet 10 — Judge answers	68

1 Field overview — the plain-language picture

1.1 Classical machine learning in healthcare

Simple. A computer is shown many past patients, each described by numbers (age, blood glucose, an image, an ECG trace) together with a known answer (“disease” / “no disease”). The computer adjusts an internal function until it reproduces the known answers, then it is used on new patients.

Technical. Supervised learning: fit a function $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ by minimising a loss on a labelled training set, then estimate generalisation on held-out data. Standard healthcare workhorses are logistic regression, support vector machines (SVM), random forests, gradient-boosted trees (XGBoost / LightGBM / CatBoost) and, for images/signals, deep neural networks.

Where it stands. On tabular clinical benchmarks these methods are *mature and often saturated*: on the Wisconsin Diagnostic Breast Cancer set, tuned RBF-SVM and gradient boosting reach $\text{AUC} \approx 0.99$ [P7]. The open problems in classical clinical ML are now about *trust* — calibration, external validation, fairness, interpretability — not raw accuracy.

1.2 Quantum computing (only what we need)

Simple. A normal bit is 0 or 1. A quantum bit (qubit) can be in a controlled blend of both at once, and a group of qubits can be *correlated* (entangled) in ways classical variables cannot. A quantum computer manipulates these blends with “gates” and then measures, getting 0/1 outcomes with tunable probabilities.

Technical. A pure state of n qubits is a unit vector in a 2^n -dimensional complex Hilbert space. Gates are unitary matrices; measurement collapses the state and returns a bitstring sampled from $|\text{amplitude}|^2$. The promise: some functions of the state are exponentially expensive to compute classically.

Where it stands (2026). We are in the *NISQ era* — Noisy Intermediate-Scale Quantum: tens to a few hundred physical qubits, no full error correction, gate error rates $\sim 10^{-3}$ – 10^{-2} , short coherence times [P1][P4][P5][P10]. Reviews note that most published health-QML hardware runs used IBM “Falcon” generation or older processors and ≤ 20 qubits [P10].

1.3 Quantum machine learning (QML)

Simple. Use a quantum computer somewhere inside a learning pipeline — usually to turn each patient’s data into a quantum state and then either (a) measure similarities between states (kernel methods) or (b) train a small quantum circuit as if it were a neural network.

Technical [P10]. A QML model takes classical data x , encodes it with a data-dependent gate $U(x)$, optionally applies trainable gates $V(\theta)$, and reads an observable:

$$f(x, \theta) = \text{Tr}[U(x, \theta) \rho_0 U^\dagger(x, \theta) \hat{O}].$$

If the x -part and θ -part separate inside the inner product, $f(x, \theta) = \langle \rho_x, \hat{O}_\theta \rangle$, the model is called a *linear quantum model*. [P10] finds that *nearly every* health-QML model in the literature (QSVM, most QNN/VQC, QCNN) is a linear quantum model — a small, well-understood corner of the theoretical design space.

1.4 Hybrid quantum–classical ML

Simple. Classical layers do the heavy lifting (cleaning, compressing, feature extraction); a small quantum circuit is dropped in as one layer; a classical optimiser trains everything.

Technical. Classical pre-processing (scaling, PCA/feature selection) → quantum encoding into few qubits → variational circuit or quantum kernel → classical readout / SVM / dense layer → classical loss + optimiser (Adam, SPSSA, L-BFGS).

Where it stands. [P5] finds $\approx 87\%$ of 94 healthcare-QML studies are hybrid; pure-quantum and quantum-inspired are $\approx 6\%$ each. [P4] builds a hybrid whose quantum layer is only **2 qubits**. Hybrid dominance is explicitly described as “a technological compromise rather than definitive superiority” [P5].

1.5 QSVM, QNN/VQC, quantum kernels, quantum feature maps, quantum optimisation — one paragraph each

Quantum feature map $U(x)$

a circuit that “writes” a data vector into a quantum state, $|\phi(x)\rangle = U(x)|0\rangle^{\otimes n}$. It plays the role of $\phi(\cdot)$ in the classical kernel trick, but the target space is the 2^n -dimensional Hilbert space [P2][P6][P9].

Quantum kernel

the similarity between two encoded states, $K(x_i, x_j) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2$, estimated by running a short circuit many times [P6][P9]. Feed this K into an ordinary SVM.

QSVM

SVM whose kernel matrix is a quantum kernel. Training and prediction are classical; only kernel entries are quantum [P6][P7][P9].

QNN / VQC

a *parameterised quantum circuit*: encode x , apply trainable rotation + entangling layers $V(\theta)$ (the “ansatz”), measure an expectation value, treat it as the prediction, and update θ with a classical optimiser using the *parameter-shift rule* for gradients [P2][P9]. “VQC” and “QNN” are used almost interchangeably in the health literature [P5].

Quantum optimisation (two senses)

(i) using a quantum device (e.g. a D-Wave annealer) to solve a QUBO that expresses an ML sub-problem — SVM training, k -means [P10]; (ii) *optimising the quantum model itself* — e.g. searching for a better feature map / kernel with a genetic algorithm (GEKO, [P6]). Sense (ii) is the one relevant to us.

1.6 Why researchers are interested

- A quantum feature map can reach a feature space that is provably hard to simulate classically — *in principle* this could separate data that no efficient classical kernel can [P3] (external: Havlíček et al. 2019; Schuld & Killoran 2019).
- Medical data is high-dimensional, heterogeneous, imbalanced and scarce — exactly the regime where people *hope* a richer representation or better data-efficiency helps [P4][P5].
- Hybrid models can be small (2–8 qubits) and therefore *runnable now* [P4][P7].

1.7 Where is the field actually standing in 2026?

THE HONEST 2026 STATUS — DO NOT OVERSELL THIS TO JUDGES

Theoretical promise: real. There exist constructed problems with a provable quantum prediction advantage [P3], and quantum feature maps are genuinely more expressive in principle [P2] (external: Havlíček 2019).

Experimental evidence on health data: *weak and inconsistent*. [P10] synthesised 16 rigorous studies and found “no consistent trend to support empirical quantum utility in digital health.” [P5] concludes QML in healthcare “remains exploratory.” [P7] pre-registered

a conservative hypothesis, found the VQC *matched but did not beat* tuned classical models, and its ZZ-kernel QSVM *underperformed badly* (AUC 0.727 vs ≥ 0.99).

Real-world clinical deployment: *none*. Zero studies in [P10] addressed health-service delivery or public health; [P5]/[P10] report almost no external validation, almost no uncertainty quantification, almost no noise characterisation, and heavy reliance on a handful of small public datasets (WDBC, PIMA, BreastMNIST).

WHY THIS MATTERS

For an SIH project this is *good news, correctly framed*. You are not entering a solved field where you must beat a world record. You are entering a field whose **own reviewers say the evaluation practice is broken**. A student team that does the evaluation *properly* — and adds one honest technical idea — is making a real, defensible contribution even if the quantum model does not win.

2 Current state of the art — a literature map

The attached papers, plus the studies they synthesise, fall into fourteen overlapping categories. For each: what people do, why, typical architecture, strengths, limitations, maturity, and relevance to us.

2.1 1. Classical ML + quantum preprocessing / quantum feature engineering

What & why. Keep a classical classifier; use a quantum circuit only to generate extra features (“Quantum Feature Engineering”, QFE) or to transform inputs. Cheap way to “add quantum”. **Architecture.** data $\rightarrow$ quantum circuit produces $\langle Z_i \rangle$ values $\rightarrow$ concatenate with classical features $\rightarrow$ XGBoost/LR. [P4] cites Safriandono et al. (2024): XGB + QFE + SMOTE–Tomek on the Indian Liver dataset, acc 81%. **Strengths.** Simple, robust, uses mature classifiers. **Limitations.** The “quantum” part is decorative; no clear evidence the quantum features add signal a classical transform could not [P5][P10]. **Maturity.** Prototyped widely, never validated. **Relevance.** A useful *ablation arm*, not a novelty.

2.2 2. QSVM / quantum kernel methods

What & why. Replace the SVM kernel with $K(x_i, x_j) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2$; hope the quantum feature space separates data a classical kernel cannot [P6][P7][P9]. **Architecture.** PCA/feature-select to n features $\rightarrow$ encode with a fixed feature map (usually the ZZ / Pauli-expansion map) on n qubits $\rightarrow$ estimate N^2 kernel entries $\rightarrow$ classical SVM. **Strengths.** Convex training; only kernel entries are quantum; theory is well developed (external: Schuld 2021 — “supervised QML models are kernel methods”). **Limitations.** Kernel estimation is $O(N^2)$ circuit evaluations; the fixed ZZ map is often a *bad* inductive bias — [P7] ZZ-QSVM AUC 0.727 vs classical ≥ 0.99 ; [P6] shows the Pauli-ZZ kernel is consistently the *worst* of all kernels tested. Exponential kernel concentration at scale (external: Thanasilp et al. 2024). **Maturity.** Most-studied QML model in health [P5][P10]; no robust advantage. **Relevance.** Baseline quantum arm + the thing our novelty (kernel/feature-map optimisation) would fix.

2.3 3. Variational Quantum Classifiers (VQC)

What & why. A quantum circuit acting like a small neural net, trained end-to-end [P7] [P9]. **Architecture.** angle-embed $x \rightarrow L$ “StronglyEntangling”/Basic entangler layers of trainable R -gates + CNOT ring $\rightarrow$ measure $\langle Z_0 \rangle \rightarrow$ sigmoid $\rightarrow$ BCE loss $\rightarrow$ Adam [P7]. [P7]: 8 qubits, 4 layers, 96 parameters. **Strengths.** Expressive; can match tuned classical models on saturated

benchmarks ([P7]: VQC internal AUC 0.992, external 0.983). **Limitations.** Barren plateaus at depth/width [P1]; extreme sensitivity to design choices — [P7] shows a Z-rotation embedding *collapses the VQC to chance* (AUC ≈ 0.50) while X/Y embeddings keep AUC > 0.98 . **Maturity.** Second most-studied health-QML model [P5]. **Relevance.** Second quantum arm; the embedding-rotation ablation from [P7] is mandatory for us.

2.4 4. Quantum Neural Networks (QNN), incl. data re-uploading

What & why. Same as VQC but the classifier idea is generalised: re-inject the data between trainable layers (*data re-uploading*) to raise expressivity with few qubits [P2]. **Architecture** [P2]. repeated blocks $L(i) = U(\theta_i + w_i \circ x)$; a *single qubit* with enough re-uploads is a universal classifier; multi-qubit + entanglement improves data-efficiency. **Strengths.** Minimal qubits; strong on toy geometric data (circle, annulus: 96–98% [P2]). **Limitations.** Local minima; benchmarked only on synthetic / tiny sets [P2]; no health validation in the source. **Maturity.** Foundational method, widely cited, rarely stress-tested clinically. **Relevance.** The right way to build our QNN arm on 2–6 qubits.

2.5 5. Hybrid (classical deep net + quantum layer)

What & why. Let dense/conv layers compress to a few numbers, hand those to a tiny quantum layer [P4]. **Architecture** [P4]. Dense(256) $\rightarrow$ Dropout $\rightarrow$ Dense(128) $\rightarrow$ Dropout $\rightarrow$ Dense(2) $\rightarrow$ *AngleEmbedding(R)* + 4 *BasicEntanglerLayers on 2 qubits* $\rightarrow$ measure $\langle Z \rangle$ $\rightarrow$ sigmoid. **Strengths.** Runs today; [P4] reports acc 84%, recall 94%, F1 87% on Indian Liver, beating the quantum baselines it cites, using only 2 qubits. **Limitations.** Quantum contribution unquantified (no ablation vs. “same net, no quantum layer”); [P4] *removed consistently-misclassified test rows* — an evaluation red flag (see Part 3). **Maturity.** The dominant practical pattern [P5]. **Relevance.** Our MVP is close to this, *but with the missing ablation added*.

2.6 6. Quantum feature maps (as an object of study)

What & why. The feature map *is* the model in kernel methods and half the model in VQCs; its design controls expressivity, trainability and kernel geometry [P2][P3][P6]. **Options.** basis, amplitude, angle/rotation encoding; Pauli/ZZ entangling maps; data re-uploading maps. **Strengths / limits.** More expressive maps $\rightarrow$ larger Hilbert space $\rightarrow$ (often) *smaller* useful kernel differences and worse trainability [P3]. **Maturity.** Actively researched; “no principled approach to link quantum algorithms to structure in classical data” [P10]. **Relevance.** Central — our novelty lives here.

2.7 7. Quantum kernel / feature-map optimisation

What & why. Stop hand-picking the map; *search* for one. [P6] (GEKO) uses a genetic algorithm over gate sequences from a hardware-native gate set $\{X, \sqrt{X}, R_z, \text{CNOT}\}$, fitness = training accuracy (“supervised”) or max normalised kernel eigenvalue (“unsupervised”). **Result** [P6]. GEKO kernels match or beat RBF and *consistently beat* the Pauli-ZZ kernel on 9 datasets incl. Parkinson’s, Drug Classification, Breast Cancer. Kernel entanglement entropy has *little correlation* with accuracy. **External:** kernel-target alignment / trainable embedding kernels (Hubregtsen et al. 2022). **Maturity.** Emerging; [P6] still only on small/UCI data, single split family, simulator. **Relevance.** *This is the most promising, least-crowded lever for us* — see Parts 4–5.

2.8 8. Quantum optimisation methods (annealing / QUBO)

What. Map SVM training, k -means, feature selection to QUBO; solve on D-Wave ($O(100\text{--}1000)$ qubits) [P10]. **Strengths.** Provable complexity advantage for *some* QUBO-SVM / balanced

k-means. **Limits.** [P10] found “no strong evidence” annealers beat gate-based or classical methods on health data. **Maturity.** Oldest health-QML thread (2018), stalled. **Relevance.** Out of scope for us (different hardware, no simulator parity).

2.9 9. Quantum-enhanced medical imaging

What. QCNN / quanvolutional layers / hybrid CNN–QSVM on MRI, CXR, histopathology, MedMNIST [P5][P9]. [P5]: imaging is the *most common* data type; QCNN used in 13 studies. **Strengths.** Classical CNN backbone carries performance; some report very high accuracy (e.g. VGG16/ResNet50 + QSVM, Alzheimer’s MRI, 99.9% — but weak baseline, single split). **Limits.** Dimensionality forces aggressive PCA/cropping before encoding, risking *train/test duplication* [P10]; quantum part rarely ablated. **Maturity.** Popular, proof-of-concept. **Relevance.** Too heavy for an SIH MVP; possible *advanced* extension.

2.10 10. Quantum-enhanced tabular health prediction

What. QSVM/VQC/hybrid on UCI-style tables: liver [P4], breast [P7][P9], Parkinson’s [P6], heart, PIMA diabetes, stroke [P5]. **Strengths.** Small qubit counts; reproducible; fair fight is feasible. **Limits.** Classical baselines saturate [P7]; datasets tiny; leakage risk high. **Maturity.** Where most “disease detection” QML sits. **Relevance.** *This is our arena.*

2.11 11. Quantum-enhanced biosignal analysis

What. ECG (MIT-BIH), EEG (BCI IV), speech/handwriting for Parkinson’s [P5]; QGAN for synthetic ECG (external: Qu 2023, in [P10]). **Strengths.** 1-D signals compress well; clear clinical framing (arrhythmia). **Limits.** Needs windowing + feature extraction; noise-model realism poor [P10]. **Maturity.** Second most common data type [P5]. **Relevance.** A strong *alternative arena* if a good open ECG/EEG early-detection task is chosen; heavier pipeline than tabular.

2.12 12. Explainable quantum ML

What. Make a QML prediction interpretable. [P7]: SHAP on a *classical surrogate* trained to mimic the VQC + a fidelity audit of the surrogate (97% label agreement, Spearman $\rho = 0.82$). **Strengths.** Lets you compare “which features drive the quantum vs classical decision.” **Limits.** Surrogate can mask local quirks; direct circuit-level attribution is “active research” [P7]. [P5]/[P10] list explainability as *underexplored*. **Maturity.** Nascent; [P7] is close to the state of the art. **Relevance.** We adopt [P7]’s recipe wholesale — it is cheap and expected.

2.13 13. Noise-aware / hardware-aware QML

What. Test the model under noise; design circuits for a device’s native gates/connectivity. [P7]: depolarising channels $p \leq 0.10$, negligible VQC AUC loss (but no T_1/T_2 , crosstalk, coherent/correlated noise). [P6]: restricts the gene pool to IBM-native gates. External: noise-induced barren plateaus (Wang et al. 2021). **Limits.** [P10]: only 2/16 rigorous studies used *any* noisy simulation; noise models were “simple” and unrealistic. **Maturity.** Recognised as essential, rarely done. **Relevance.** A cheap, high-credibility component of our evaluation (Part 17).

2.14 14. Robustness and generalization

What. External validation on an independent cohort; internal–external cross-validation across strata; learning curves; feature-selection stability; calibration; decision-curve analysis; multiple seeds. [P7] does *all* of these and is the template. [P3] adds the theory: prediction-error bounds, the *geometry test*, and why more qubits can make things worse. **Limits.** Almost no other

health-QML paper does this [P10]. **Maturity.** [P7] (Jun 2026) is essentially the current frontier. **Relevance.** *Our evaluation section should look like [P7]’s.*

READING THE MAP

Categories 2–7 and 10–14 are our working space. Category 7 (**feature-map / kernel optimisation**, [P6]) is the *least crowded* technical lever. Categories 12–14 (explainability + robustness + noise, embodied in [P7]) are *expected hygiene* we must simply execute. Categories 8–9 and 11 are out of scope for an MVP.

3 Literature gap analysis

This section is deliberately hard-nosed. For each candidate gap we state the evidence, why it matters, what has already been tried, what is still missing, difficulty for a student team, and SIH potential. The final column of the table also assigns a status:

A already reasonably solved / addressed — do not claim as novelty.

B partially addressed — room for a contribution but crowded.

C substantially open — realistic target.

D open but too hard / out of scope for an SIH prototype.

3.1 The gap table

Gap	Evidence	Why it matters	Existing attempts	Still missing	Status
No consistent quantum advantage on health data	[P10] “no consistent trend”; [P5] “exploratory”; [P7] pre-registered null, confirmed	If there is no advantage, the whole pitch must be reframed around method, not performance	[P3] geometry test explains <i>why</i> ; [P7] shows honest parity	A clean, public, leakage-free head-to-head on an <i>early-detection</i> task with strong baselines	B / high (as)
Weak classical baselines	[P5]: quantum often compared only to LR / default RF; [P7] & external (Bowles et al. 2024) call this out	A “win” over a weak baseline is meaningless to judges	[P7]: Optuna-tuned RBF-SVM + 3 boosting libs as baseline	Community habit still weak; a well-baselined <i>early-detection</i> study is still uncommon	
Unrealistic benchmarking / no scaling study	[P10]: only 1/16 studies plotted performance vs. problem size	Advantage claims require showing the trend, not one lucky point	[P3]: 30-qubit scaling on engineered data (not health)	Any small <i>systematic</i> qubit-count / feature-count sweep on health data	C
Ideal-simulator dependence	[P10]: 131/138 sim-only studies used <i>ideal</i> sims; [P5] same	A model that only works noiseless is not a result	[P7]: depolarising sweep; [P6]: native gate set	Realistic noise (T_1/T_2 , readout, crosstalk, coherent) + mitigation on a health task	C
Quantum hardware noise / NBP	[P1] barren plateaus; external Wang 2021 noise-induced BP; [P4][P5] list decoherence	Determines whether anything survives on real QPUs	Shallow circuits, structured init, hardware-efficient ansatz	Task-specific noise budget for a <i>tabular early-detection</i> circuit	C (test) /
Scalability of data encoding	[P10] Table 1: amplitude enc. $O(\log d)$ qubits but $O(Nd)$ depth; [P8] amplitude-encodes genomic PCA features	Encoding cost can erase any speed-up	QRAM (not near-term), parallel loaders (special hardware), QRAC for discrete vars	A fair-cost comparison that <i>counts</i> encoding depth in the “quantum budget”	D (general) /

Gap	Evidence	Why it matters	Existing attempts	Still missing	St.
Small datasets	[P5][P10]: WDBC ($n=569$), PIMA, BreastMNIST reused everywhere; [P4] $n=583$	Tiny $n \Rightarrow$ unstable estimates, easy overfitting	[P7] learning curves + bootstrap CIs	Multi-dataset / multi-cohort evaluation on the <i>same disease</i>	B
Class imbalance	[P4] 416:167; [P5] lists imbalance as underexplored; [P2][P4] claim QML needs less balancing	Early detection = rare positives; accuracy is misleading	[P4]: SMOTE, focal loss, class weights (SMOTE hurt recall $\rightarrow 76\%$)	Whether a quantum model is <i>genuinely</i> more imbalance-robust, tested properly (PR-AUC, no leakage from resampling)	
Data leakage	[P4] removed hard-to-classify test rows post-hoc; [P10]: PCA-before-split, image de-dup after reduction	Leakage inflates every number; kills credibility	[P7]: every fit-step inside CV folds only; [P4]: scaler fit on train folds only	A <i>leakage audit</i> as an explicit deliverable is rare	
Lack of external validation	[P10]: only 1 study used real EHR; [P5]: “limited”	Internal AUC $\neq$ deployable model	[P7]: WDBC $\rightarrow$ WBC method-level external validation	External validation for <i>early-detection tabular</i> tasks; multi-centre	
No uncertainty estimation	[P10]: only 1/16 reported error bars; [P5] lists it as a gap	Clinicians need calibrated confidence, not a point label	[P7]: bootstrap CIs, Brier, reliability diagrams, DeLong test	Predictive uncertainty for quantum outputs (shot noise + epistemic) surfaced to a user	
No robustness testing	[P10] Table 3 “sensitivity analysis: typically untested”	A model that breaks under a 1% perturbation is unsafe	[P7]: ablation, noise, learning curve, feature stability, IECV	Same battery applied to an <i>early-detection</i> pipeline + adversarial-style feature perturbation	
Poor reproducibility	[P10]: most studies gave no code/data; [P5]: heterogeneous protocols	Can’t build on what you can’t re-run	[P4][P6][P7] re-release code	Reproducibility is a <i>choice</i> — ours by default	
Fixed feature-map selection	[P6]: manual design “poses significant challenges”; [P7]: ZZ map “not well aligned”	The map <i>is</i> the model; a bad default guarantees failure	[P6] GEKO (genetic); external: kernel-target alignment, AutoQML	Map optimisation <i>plus</i> leakage-free, externally-validated, noise-checked evaluation on <i>early health detection</i> ; principled search cost	C / high
QNN trainability / barren plateaus	[P1] (exp. vanishing gradients in n); [P10]: no study characterised its landscape	Untrainable = no model	Shallow / structured ansatz, data re-uploading [P2], layerwise training	An <i>empirical</i> landscape + gradient-variance report for our specific circuit — almost nobody does this in health QML	C (measure) /
Circuit depth / hardware limits	[P4][P5]: depth + qubit constraints force 2–10 qubits	Deep circuits don’t run and don’t train	Keep ≤ 8 qubits, ≤ 4 –6 layers [P4][P7]	n/a — accept the constraint	
Explainability / clinical interpretability	[P5][P10]: underexplored; [P7] is close to SoTA	No explanation $\Rightarrow$ no clinician trust, no deployment	[P7]: surrogate SHAP + fidelity audit; classical SHAP reference	Direct circuit-level attribution; explanation <i>stability</i> under noise	B / medium (adopt)

Gap	Evidence	Why it matters	Existing attempts	Still missing	St.
Generalisation across datasets / populations	[P5][P10]: single-source datasets; [P7] cross-representation only	Fairness + trans-portability	[P7]: WDBC→WBC; IECV across radius strata	True multi-population / multi-site early-detection generalisation	C /
Computational cost honesty	[P5]: “report training time, circuit depth, shots, qubits”; [P10]: parallelism assumptions unstated	Hidden cost invalidates “efficiency” claims	[P7] Table 3 (wall-clock, per-circuit ms)	A cost ledger as a first-class result	B
Lack of standardised evaluation	[P10] Table 3 proposes a framework; [P5] calls for one	Meta-analysis impossible without it	[P10] minimal-requirements table; [P7] TRIPOD+AI	Adoption. We can <i>adopt</i> [P10]+TRIPOD+AI and show it works for a student team	B / hi

3.2 Which gaps are already being addressed by recent papers?

GAPS THAT ARE LARGELY CLAIMED — DO NOT PRESENT AS YOUR NOVELTY

- **“Rigorous, externally validated, explainable QML benchmarking in the NISQ era”** — done by Zorlu & Colak 2026 [P7] (*Diagnostics* 16:1996) for breast cancer. They already provide: tuned saturating baseline, internal–external CV, external cohort, ablation, noise, learning curves, feature stability, decision-curve analysis, dual SHAP + surrogate fidelity audit, TRIPOD+AI reporting. If you propose “do rigorous benchmarking” as your contribution, a judge who knows this paper will say “already done.” *Your move: apply the same rigour to a different, non-saturated, early-detection task and add a technical lever on top.*
- **“Optimise the quantum kernel / feature map with a genetic algorithm”** — done in general by Creevey et al. 2026 [P6] (GEKO) and earlier by Altares-López et al. 2021 and AutoQML (Altares-López et al. 2022). External: *kernel-target alignment* for trainable embedding kernels (Hubregtsen et al., *Phys. Rev. A* 106:042431, 2022, arXiv:2105.02276); *Training quantum kernels* (Gentinetta et al. 2023). *Your move: not “we invented kernel search” — rather “we combine kernel/feature-map search with a leakage-free, externally-validated, noise-tested, imbalance-aware evaluation on an early-detection task, and we report the search’s own cost and overfitting risk,” which [P6] does not.*
- **“Data re-uploading makes small circuits expressive”** — Pérez-Salinas et al. 2020 [P2]. Established; use it, cite it, don’t claim it.
- **“Barren plateaus / trainability theory”** — McClean et al. 2018 [P1]; extensive follow-up (Cerezo et al. 2021; Larocca et al. 2024 review, arXiv:2405.00781). Use the design rules; don’t claim to solve it.
- **“Classical ML with data can match quantum models; here is how to test for advantage”** — Huang et al. 2021 [P3]. The *geometry test* is a tool you should *use*, not reinvent.
- **“Benchmarking QML is subtle / baselines are too weak”** — external: Bowles, Ahmed, Schuld, “Better than classical? The subtle art of benchmarking quantum machine learning models,” arXiv:2403.07059 (2024). Cite it as your methodological backbone.

GAPS THAT ARE STILL MEANINGFULLY OPEN FOR A STUDENT TEAM

1. **Feature-map / kernel optimisation evaluated *honestly* on an early-detection task** — with leakage-free CV, an independent cohort, imbalance handled without leakage, noise robustness, *and* an explicit account of the search’s overfitting risk and compute cost. [P6] optimises; it does *not* do the clinical-grade evaluation. [P7] evaluates; it uses a *fixed* ZZ kernel that it admits is badly aligned. **Nobody has joined the two on early health detection.** (Status C, our #1.)
2. **Imbalance-robustness of quantum vs classical models, tested correctly** — is the folk claim ([P2][P4]: “QML needs less balancing”) actually true under PR-AUC, leakage-free resampling, and multiple seeds? Currently asserted, never cleanly measured. (Status C, our #2.)
3. **Predictive-uncertainty surfacing for quantum outputs** — combine shot-noise uncertainty with an ensemble/bootstrap epistemic estimate and show calibration. Only 1/16 rigorous studies reported *any* error bar [P10]. (Status C, supporting component.)
4. **An empirical trainability / gradient-variance report for the exact circuit used on a health task** — [P10] notes *no* study did this. Cheap to add, high credibility. (Status C, supporting component.)

4 What should our novelty be? — candidate directions

Five candidate directions, each scored against the same 12 questions. Then ranked.

4.1 Direction A — Leakage-free, externally-validated evaluation of an *optimised* quantum feature map / kernel for early health detection

1. **Existing problem.** QSVM/VQC results on health data use a *fixed* feature map (almost always ZZ / Pauli), which is often a poor inductive bias [P7][P6].
2. **Research gap.** [P6] optimises the map (genetic algorithm) but only on UCI/toy data, one split family, ideal simulator, no external cohort, no imbalance analysis, no reporting of the search’s own overfitting/compute cost. [P7] evaluates rigorously but with a fixed, admittedly mis-aligned kernel. Nobody joins “optimise the map” + “evaluate it to clinical-benchmark standard” on an *early-detection* task.
3. **Proposed solution.** A pipeline that (a) searches a small space of feature maps / kernels (kernel-target alignment first; genetic search as the advanced option), (b) does the search *strictly inside* cross-validation folds (nested CV) so map selection cannot leak, (c) compares the resulting QSVM/VQC against Optuna-tuned RBF-SVM + gradient boosting under an identical protocol, (d) validates on an independent cohort, (e) tests imbalance, noise and seeds, (f) reports the search’s compute budget and a “did the search overfit?” check.
4. **Why different.** The novelty is not the optimiser; it is the *honest integration*: optimised map + nested-CV leakage control + external cohort + imbalance/noise battery + search-cost ledger, on early health detection. That specific combination does not exist in [P1]–[P10] or the external work we found.
5. **What previous papers already did.** [P6]: genetic kernel search (toy/UCI, ideal). [P7]: rigorous eval (fixed kernel, breast cancer). External: kernel-target alignment (Hubregtsen 2022); AutoQML (Altares-López 2022); benchmarking critique (Bowles 2024).
6. **What remains different.** The join, on an early-detection task, with a leakage audit and a search-cost ledger, and an explicit test of whether map optimisation actually beats the best classical kernel *when the classical side is also tuned*.
7. **Technical difficulty.** Medium. Kernel-target alignment is ~30 lines on top of a QSVM.

Genetic search (GEKO-style) is a weekend of engineering. Nested CV is standard.

8. **Required data.** One early-detection tabular dataset with a natural second cohort (e.g. two diabetes cohorts; two CKD cohorts; PhysioNet early-sepsis; two heart-failure registries). Public, $n \gtrsim 500$, imbalanced.
9. **Required quantum resources.** 4–8 qubits, depth ≤ 8 , statevector + noisy simulator; optional 1 short IBM/IonQ free-tier run for the final kernel.
10. **Expected experimental setup.** 3 classical baselines vs. ZZ-QSVM vs. alignment-optimised QSVM vs. genetic-optimised QSVM vs. data-reuploading VQC; nested 5×5 CV; external cohort; PR-AUC + AUC + calibration; noise sweep; 10 seeds; ablations.
11. **Risk of duplication.** Low–medium. The pieces exist separately; the specific integration on early detection does not. Cite [P6][P7] up front to pre-empt “already done.”
12. **SIH feasibility.** High. Every component is a known recipe; the contribution is discipline + integration + one honest technical lever.

4.2 Direction B — Is a quantum model genuinely more robust to class imbalance? A clean test

Existing problem: [P2][P4] *claim* QML needs less class balancing; never cleanly shown. Gap: no leakage-free, PR-AUC-based, multi-seed comparison of quantum vs classical degradation as the positive rate drops. Proposed: fix everything else; sweep the training imbalance ratio ($1:1 \rightarrow 1:20$) with and without resampling (resampling done inside folds only); measure PR-AUC, recall at fixed precision, calibration, for tuned classical vs QSVM vs VQC. Difficulty: low–medium. Data: any imbalanced early-detection table. Quantum: 4–8 qubits. Duplication risk: low (folk claim, untested). SIH feasibility: high; very “experiment-shaped,” easy to present. Weakness: if the answer is “no difference” (likely per [P3][P7]), the story is a *negative result* — scientifically valuable, less flashy.

4.3 Direction C — Uncertainty-aware quantum early-detection with a calibrated risk output

Existing problem: [P10] — almost no QML study reports uncertainty; clinicians need calibrated confidence. Gap: combine *aleatoric* shot-noise uncertainty (finite measurement counts) with *epistemic* uncertainty (bootstrap/deep-ensemble of circuits) into one calibrated probability + interval, and show reliability diagrams / ECE beat a single-point model. Proposed: ensemble of k VQCs (different seeds / re-uploading depths) + shot-count propagation + temperature scaling on a held-out calibration split; deliver “risk 0.71 [0.58–0.83], calibrated.” Difficulty: medium. Duplication risk: low. SIH feasibility: medium–high; strong dashboard story. Weakness: uncertainty methods are mostly classical; the “quantum” angle is thin unless paired with A or B.

4.4 Direction D — Noise-budget characterisation for a tabular early-detection circuit

Existing problem: [P10] — 131/138 sim-only studies used ideal simulators; [P1]/Wang 2021 — noise kills trainability. Gap: a task-specific “how much noise can this early-detection circuit tolerate, and does error mitigation recover it” study — almost nobody does this for health tables. Proposed: realistic noise model (depolarising + readout + thermal relaxation + a coherent term), sweep error rates, apply zero-noise extrapolation / measurement-error mitigation, report AUC/PR-AUC vs. noise with and without mitigation. Difficulty: medium (noise modelling is fiddly). Duplication risk: low–medium. SIH feasibility: medium. Weakness: it is an evaluation study, not a new model — best as a *component* of A.

4.5 Direction E — Quantum-enhanced early detection on a biosignal (ECG/EEG) with a classical feature extractor + optimised quantum head

Existing problem: imaging/biosignal QML uses fixed maps and weak baselines [P5][P9]. Gap: same as A but on a signal. Proposed: classical feature extraction (HRV / spectral / wavelet) → optimised quantum kernel head → rigorous eval. Difficulty: medium-high (signal pipeline + windowing + leakage across windows of the same patient is a real trap). Duplication risk: low. SIH feasibility: medium. Weakness: patient-level leakage in biosignals is easy to get wrong and hard to defend; heavier than tabular.

4.6 Ranking

RANKED DIRECTIONS

#1 — Most promising: Direction A (optimised feature map / kernel, evaluated honestly on early health detection). Technically meaningful, defensible from [P6]+[P7], implementable, testable, directly about early detection, demonstrable in a dashboard. It also *absorbs* Directions C and D as sub-components (uncertainty output; noise budget) and B as one experiment (imbalance sweep).

#2 — Backup: Direction B (imbalance-robustness clean test). Smallest, safest, most clearly “an experiment”; good if the team is small or time-poor. Reframe as “we test a widely-repeated claim.”

#3 — Optional: Direction C (calibrated uncertainty output) as a standalone if the team wants a product/dashboard emphasis over a research emphasis.

SIH JUDGE NOTE

Do *not* pick a direction because it “sounds like quantum magic.” Direction A is chosen because: (i) it targets a gap two 2026 papers leave open ([P6] optimises but doesn’t validate; [P7] validates but doesn’t optimise); (ii) every sub-step is a known, teachable recipe; (iii) it produces a real result *whichever way it comes out* — either “optimising the map closes the gap to classical” or “even an optimised quantum kernel does not beat a tuned classical kernel on this task,” and both are publishable, honest findings.

5 Our proposed research idea

5.1 The exact problem we are solving that existing systems do not

Existing quantum classifiers for disease detection use a **fixed, hand-chosen quantum feature map** (nearly always the ZZ / Pauli map). When that map is a poor match to the data geometry, the quantum model *underperforms even a modestly-tuned classical model* ([P7]: ZZ-QSVM AUC 0.727 vs classical ≥ 0.99 ; [P6]: Pauli-ZZ is the worst kernel on every dataset tested). Conversely, the one line of work that *does* adapt the map to the data — genetic / alignment kernel optimisation [P6] — has only been shown on toy and UCI datasets, on a single train/val/test split family, on an *ideal* simulator, with **no leakage control for the selection step, no external cohort, no imbalance analysis, no noise test, and no accounting of the search’s own compute cost or overfitting risk**. So the field does not currently know: *when a quantum feature map is adapted to the data under a leakage-free, clinically-styled evaluation, does it close (or open) the gap to a well-tuned classical model on an early-detection task?*

5.2 Research Problem

Design and rigorously evaluate a hybrid quantum–classical early-detection classifier whose quantum feature map is *selected from data* (not fixed), under an evaluation protocol strong enough that any performance claim — positive or null — is trustworthy.

5.3 Research Gap

The intersection of $\{\text{data-adaptive quantum feature map / kernel}\} \cap \{\text{leakage-free nested selection}\} \cap \{\text{independent-cohort external validation}\} \cap \{\text{imbalance + noise + seed robustness}\} \cap \{\text{search-cost \& overfitting ledger}\} \cap \{\text{early health detection}\}$ is empty in [P1]–[P10] and in the external literature we surveyed ([P6] covers the first element; [P7] covers elements 2–4; nobody covers all).

5.4 Hypothesis

H1. A data-adaptive quantum feature map (kernel-target-alignment optimised; genetic search as the advanced variant) yields a QSVM whose leakage-free cross-validated PR-AUC is *significantly higher* than the fixed ZZ-map QSVM on the chosen early-detection task.

H2 (the hard one). The adaptive-map QSVM’s PR-AUC is *at least statistically non-inferior* to the best of {Optuna-tuned RBF-SVM, XGBoost, LightGBM} under the identical protocol, on the internal test set *and* the external cohort.

H3. The adaptive-map model degrades *no faster* than the tuned classical baseline as (a) training imbalance increases and (b) simulated hardware noise increases.

5.5 Proposed Contribution

1. A reusable, open, **leakage-audited** hybrid pipeline for early health detection with a data-adaptive quantum feature map, packaged so any team can rerun it on a new dataset.
2. The first evaluation of quantum feature-map optimisation on an early-detection task that includes external-cohort validation, PR-AUC-based imbalance analysis, a realistic noise sweep with mitigation, multi-seed statistics, and a **search-cost + overfitting ledger**.
3. An honest verdict on H2/H3 for that task, reported *whichever way it comes out*, plus a calibrated, explainable risk output (SHAP + surrogate-fidelity audit per [P7]; bootstrap + shot-noise uncertainty per Direction C).
4. A short “*when is an adaptive quantum map worth it?*” decision note tying our result back to the [P3] geometry test.

5.6 Expected Benefit

- If H2 holds: evidence that *map adaptation*, not raw “quantumness,” is what a near-term quantum classifier needs to be competitive on health tables — a concrete, actionable finding for the field.
- If H2 fails: a clean, well-controlled data point that even an optimised quantum kernel does not beat a tuned classical kernel on early detection — directly strengthening the [P3]/[P10] picture and saving other teams the effort.
- Either way: a template pipeline + a risk dashboard that a hospital data-science team could actually inspect.

5.7 Limitations (state these before a judge does)

- Simulation-first; at most one short real-QPU run. Not hardware-clinical-ready.
- Small qubit counts (4–8) $\Rightarrow$ we cannot probe the regime where a quantum advantage is theoretically argued to appear.

- Public datasets; “external cohort” is method-level, likely same research lineage (as in [P7]) — not multi-centre.
- Kernel estimation is $O(N^2)$ circuits; we cap N for tractability and *report* that as a cost, not hide it.
- Genetic search can overfit the validation signal; we mitigate with nested CV + a held-out confirmation set, but residual risk remains and we quantify it.

5.8 What would falsify our hypothesis?

FALSIFICATION CRITERIA — WRITTEN IN ADVANCE, REPORTED HONESTLY

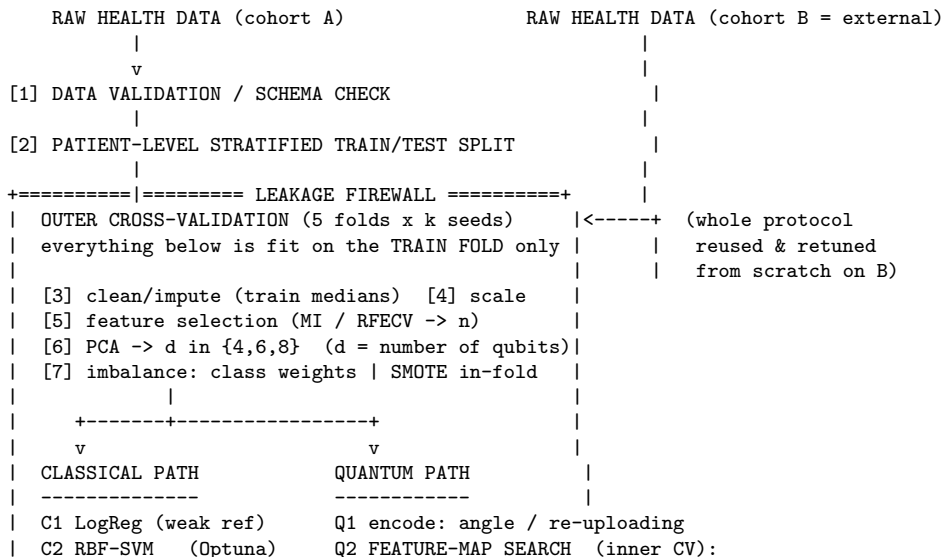
- **H1 falsified** if the adaptive-map QSVM’s nested-CV PR-AUC is not significantly above the ZZ-map QSVM’s (paired test across folds/seeds, $\alpha = 0.05$). Then: “map adaptation did not help here” — we report it and analyse why (geometry test, kernel concentration).
- **H2 falsified** if, under leakage-free evaluation on the internal test set *and* the external cohort, the best tuned classical model’s PR-AUC exceeds the adaptive-map model’s by a margin whose 95% CI excludes zero (DeLong / bootstrap). Then: **we will state plainly that we found no quantum advantage**, and the project’s contribution becomes the pipeline + the negative result + the decision note — exactly as [P7] did for breast cancer.
- **H3 falsified** if the quantum model’s PR-AUC drops significantly faster than the classical baseline’s under the imbalance sweep or the noise sweep. Then: “the folk claim that QML is more imbalance/noise robust does not hold for this task.”

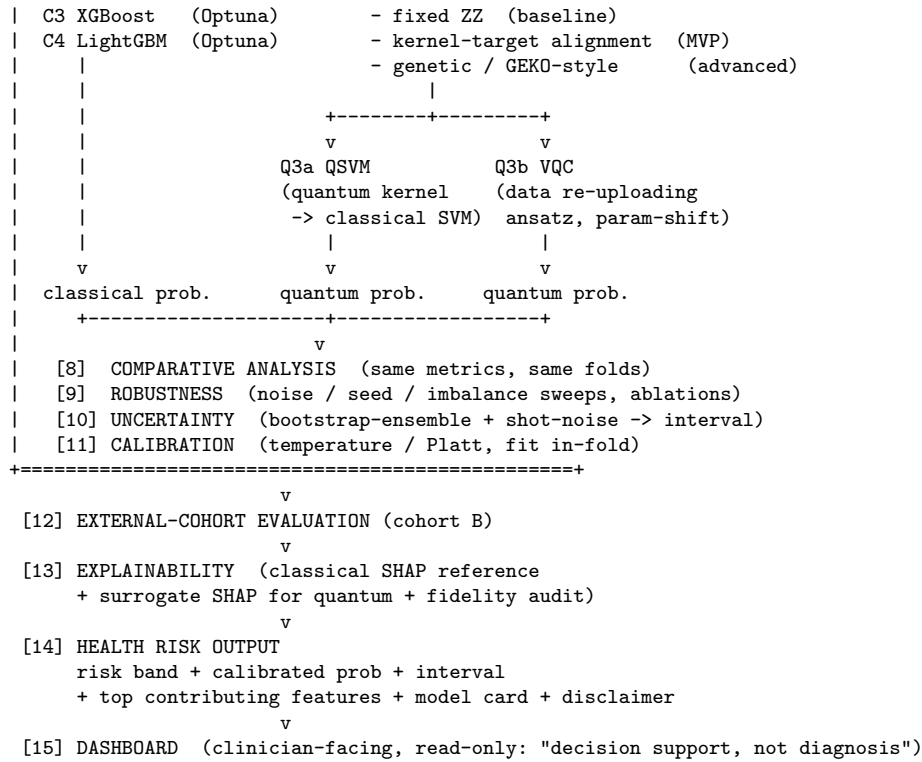
We commit to publishing the falsifying result rather than tuning until the quantum model wins.

6 Proposed architecture

The example architecture in the brief is a good skeleton. We modify it in four ways dictated by the gap analysis: (1) all fitting steps (scaler, feature selection, PCA, *feature-map selection*, resampling, calibration) move *inside* the cross-validation loop — a “leakage firewall”; (2) the quantum path carries a *feature-map optimiser*, not a fixed map; (3) an explicit *external-cohort* branch; (4) an *uncertainty* block feeding the risk output.

6.1 Architecture diagram





6.2 Every block explained

Block	In	What happens	Out	Why required (paper)	Tech	
1 Validation	raw CSV	type/range/units check, missing-rate report, target sanity	clean frame	typed	[P8] shows what happens without it (hand-waved data); [P10] reproducibility	pandas, pandera
2 Split	typed frame	<i>patient-level</i> stratified hold-out (never split rows of one patient)	train / test		[P4] leakage from post-hoc row removal; [P10]	scikit-learn
3 Clean	train fold	impute with <i>train-fold</i> medians; drop exact dups	imputed fold		[P4] (mean impute A/G ratio); [P10]	scikit-learn
4 Scale	imputed fold	fit robust/standard scaler on train fold only	scaled fold		[P4][P7] (scaler fit on train only); encoding needs bounded ranges	scikit-learn
5 Feature selection	scaled fold	mutual-info / RFECV, keep top n	n features		[P6][P7] select before encoding; [P10] impact “unaddressed”	scikit-learn
6 PCA	n features	PCA to $d \in \{4, 6, 8\}$ = qubit budget; fit on train fold	d comps		[P4][P6][P8] PCA-then-encode; [P10] warns of post-PCA duplication	scikit-learn

Block	In	What happens	Out	Why required (paper)	Tech
7 Imbalance	d comps	class weights by default; SMOTE <i>inside</i> fold as an arm	balanced fold	[P4] SMOTE hurt recall; [P5] imbalance under-explored	imbalanced-learn
C1–C4 Classical	d comps	fit LogReg / Optuna-tuned RBF-SVM, XG-Boost, LightGBM	class probs	[P7]: saturating tuned baseline is mandatory; [P5]: weak baselines invalidate claims	scikit-learn, xgboost, lightgbm, optuna
Q1 Encoding	d comps	map each comp to a rotation angle; optional re-uploading	$ \phi(x)\rangle$	[P2] re-uploading; [P10] Table 1 encoding costs	PennyLane / Qiskit
Q2 Feature-map search	train fold	<i>inner</i> CV: ZZ baseline; kernel-target alignment (MVP); genetic GEKO-style (advanced)	selected map	[P6] (search); [P7] (ZZ misaligned); nested CV stops leakage	PennyLane, DEAP
Q3a QSVM	selected map	estimate kernel matrix $K_{ij} = \langle \phi(x_i) \phi(x_j) \rangle ^2$; classical SVM on K	class probs	[P6][P7][P9]	Qiskit FidelityQuantumKernel / PennyLane
Q3b VQC	encoded x	data re-uploading ansatz, train θ with parameter-shift + Adam	class probs	[P2][P7][P9]; [P1] design rules	PennyLane
8 Comparative	all probs	identical metrics on identical folds: AUC, PR-AUC, recall@precision, Brier, MCC	metric table	[P7]; [P10] standardised eval	scikit-learn
9 Robustness	trained models	noise sweep, seed sweep, imbalance sweep, ablations	robustness curves	[P7]; [P10] Table 3 sensitivity	PennyLane noisy sim
10 Uncertainty	ensemble/bootstrap + shots	epistemic interval + aleatoric shot-noise band	prob $\pm$ interval	[P10] (1/16 studies had error bars)	numpy
11 Calibration	probs + calib. split	temperature / Platt scaling fit in-fold	calibrated probs	[P7] (reliability, Brier); TRI-POD+AI	scikit-learn
12 External cohort	cohort B	rerun <i>whole</i> protocol from scratch; evaluate	external metrics	[P7] method-level external validation; [P10]	—
13 Explainability	best classical + quantum	TreeSHAP (reference); surrogate-model SHAP for quantum + fidelity audit	feature attributions	[P7]; [P5][P10] gap	shap
14 Risk output	calibrated prob + interval + SHAP	risk band, prob, interval, top-3 features, model card, disclaimer	structured JSON	[P7] (responsible reporting)	pydantic

Block	In	What happens	Out	Why required (paper)	Tech
15 Dash-board	risk JSON	read-only clinician view; “decision support, not diagnosis”	web page	[P7][P10] (clinical framing)	FastAPI + minimal frontend

WHY THIS SHAPE AND NOT THE BRIEF'S DEFAULT

The brief’s skeleton puts preprocessing / PCA *above* the classical/quantum split as one-time steps. That is the #1 leakage mechanism in health QML [P4][P10]. Our version wraps *all* data-dependent fitting — including feature-map selection — inside the CV loop. We also refuse to “force QSVM + QNN + GEKO together”: the QSVM is the primary quantum model (kernel search is where our lever lives); the VQC is a second arm for comparison; GEKO-style genetic search is an *advanced* setting of block Q2, off by default in the MVP (kernel-target alignment is the MVP optimiser).

7 Classical ML foundation (only what the prototype needs)

Format for each concept: **Simple** → **Technical** → **Healthcare example** → **Why we need it** → **Mistake to avoid** → **Pipeline location**.

7.1 Data objects

Dataset	S: a table — rows = patients, columns = measurements + one answer. T: $\{(x_i, y_i)\}_{i=1}^N$, $x_i \in \mathbb{R}^p$. H: 583 liver patients, 10 markers [P4]. W: the model can only learn what the data contains. M: assuming the sample represents the deployment population. P: block 1–2.
Features	S: the input columns (age, glucose, bilirubin). T: components of x_i ; may be numeric, ordinal, categorical. H: WDBC = 30 nuclear-morphology numbers [P7]. W: feature quality caps model quality. M: leaving an ID or a post-diagnosis field in x (leakage). P: block 5.
Labels	S: the answer column. T: y_i ; for us $y_i \in \{0, 1\}$. H: 1 = disease, 0 = healthy [P4]. W: supervised learning needs them. M: label noise / inconsistent labelling; not checking class balance. P: block 2.
Classification	S: predict a category. T: learn $P(y=1 x)$. H: disease vs. no disease. W: early detection is a classification (or risk-scoring) task. M: treating a probability as a hard label too early. P: blocks C1–C4, Q3.
Binary classification	S: exactly two classes. T: threshold $P(y=1 x)$ at τ . H: all our tasks. W: simplest, matches most early-detection framing. M: using $\tau = 0.5$ blindly when the disease is rare or a miss is costly. P: block 14.

7.2 Preparing data

Data cleaning	S: fix obvious errors. T: range checks, de-dup, unit harmonisation. H: [P4] de-duplicated 13 rows. W: garbage in, garbage out. M: cleaning using information from the whole dataset. P: block 3.
Missing values	S: some cells are empty. T: impute (mean/median/KNN) or model-aware handling. H: [P4] imputed 4 A/G-ratio values with the <i>training</i> mean; [P7] used class-conditional medians for “bare nuclei.” W: most models can’t take NaN; encoders definitely can’t. M: imputing before the train/test split ⇒ leakage. P: block 3.
Encoding (categorical)	S: turn text categories into numbers. T: one-hot / ordinal / target encoding. H: [P4] mapped gender F/M → 0/1. W: math needs numbers. M: target-encoding with test labels visible. P: block 3.

Scaling / Normalisation / Standardisation **S**: put all features on a comparable size. **T**: *normalise* = rescale to $[0, 1]$ (min-max); *standardise* = subtract mean, divide by SD (*z*-score); *robust scale* = subtract median, divide by IQR ([P7]). **H**: bilirubin 0–50 vs. albumin 2–5 — without scaling, bilirubin dominates. **W**: SVMs, PCA, gradient descent and *quantum angle encoding* are all scale-sensitive; angles must land in a bounded range ($[0, \pi]$ in [P7]; $[-\pi/2, \pi/2]$ in [P6]). **M**: fitting the scaler on train+test together. **P**: block 4 (and directly feeds Q1).

7.3 Choosing and shrinking features

Feature selection. **S**: keep only the useful columns. **T**: filter (mutual information, χ^2), wrapper (RFECV — recursively drop the weakest feature, cross-validate), or embedded (L1, tree importance). **H**: [P7] used RFECV + a mutual-information tie-break to land on exactly 8 features (= its qubit budget). **W**: we can only encode a few features into few qubits; fewer, better features = shorter circuits, less overfitting. **M**: selecting features on the full dataset then cross-validating — the CV score is then optimistic (leakage). **P**: block 5, *inside* the CV loop.

PCA (Principal Component Analysis). **S**: rotate the data so the first few new axes capture most of the spread, then keep those. **T**: eigen-decompose the covariance matrix; project onto the top- d eigenvectors; the top- d capture a chosen % of variance. **H**: [P6] used PCA to hit 95% variance — WDBC 30 $\rightarrow$ 10 comps, Parkinson’s 22 $\rightarrow$ 8; [P8] PCA’d genomic data before amplitude encoding. **W**: maps an arbitrary feature count onto a fixed, small qubit count; decorrelates features (helps some encoders). **M**: (i) PCA before the split; (ii) PCA so aggressive that two different patients collapse to the same point — [P10] warns this creates *train/test duplication* and fake accuracy; (iii) forgetting PCA is a fitted transform (fit on train fold only). **P**: block 6.

7.4 Evaluating honestly

Train / validation / test split. **S**: learn on one part, tune on a second, judge *once* on a third. **T**: e.g. 60/20/20; the test set is touched exactly once, at the end. **H**: [P4] 75/25; [P7] 70/30 with stratification; [P6] 48/32/20. **W**: the test set estimates deployment performance; if you tune on it, that estimate is a lie. **M**: “peeking” — choosing a model because it scored well on the test set. **P**: block 2.

Cross-validation (CV). **S**: rotate which fold is the validation set, average the scores. **T**: k -fold: split train into k parts, train on $k-1$, validate on 1, repeat k times; *stratified* keeps class ratios; *nested* CV has an inner loop for model/hyper-parameter/*feature-map* selection and an outer loop for unbiased scoring. **H**: [P4][P7] used stratified 5-fold; [P7] added leave-one-radius-quartile-out (internal-external CV). **W**: small health datasets give noisy single-split scores; CV averages that out. **Nested** CV is what makes our feature-map search leakage-free. **M**: doing feature selection / SMOTE / map search *outside* the CV loop. **P**: the whole outer box in block 6–11.

DATA LEAKAGE — THE SINGLE MOST IMPORTANT CONCEPT IN THIS DOCUMENT

S: the model saw something at training time it will not have at prediction time — so its scores are fake. **T**: any statistic computed using test/validation data that then influences training: scaler means, imputation values, feature-selection choices, PCA axes, resampling, hyper-parameter choice, *feature-map selection*, threshold selection. **H (real, from our sources)**: [P4] *removed* test rows that were “consistently misclassified across all 10 executions” — i.e. used model performance on the test set to edit the test set; [P10] flags PCA-before-split and post-reduction image de-duplication across the field. **W**: leaked results do not reproduce, mislead judges, and would be dangerous in a clinic. **M**: the fix is the *leakage firewall* — fit every data-dependent step inside the current training fold only, then apply unchanged to held-out data ([P7]’s explicit protocol). **P**: the firewall wraps blocks 3–11. *We will run and*

publish a leakage audit as a deliverable.

Class imbalance. **S:** far more of one class (usually many healthy, few sick). **T:** handle with class weights, focal loss, resampling (SMOTE / undersampling), or threshold tuning; evaluate with PR-AUC / recall-at-precision, *not* accuracy. **H:** [P4] 416:167, used SMOTE + focal loss + smoothed class weights — and found SMOTE *cut recall* from 94% to 76%. **W:** early detection = rare positives; a 95%-accurate model that never flags the disease is useless. **M:** (i) accuracy as the headline metric; (ii) SMOTE applied *before* the split or *outside* CV folds — synthetic points leak; (iii) resampling the test set. **P:** block 7 (in-fold) + block 8 metric choice.

7.5 The models we will actually use

Logistic Regression (LR). **S:** weighted sum of features $\rightarrow$ squashed to a probability. **T:** $P(y=1 | x) = \sigma(w^\top x + b)$; fit by maximum likelihood; L1/L2 regularised. **H:** the standard first clinical risk model; [P7] uses it as the RFECV base learner. **W:** our *weak reference* baseline — if quantum can't beat LR, stop. **M:** presenting LR as a *strong* baseline (it is not; see [P5]'s critique). **P:** block C1.

Support Vector Machine (SVM). **S:** draw the widest possible “street” between the two classes; the edge patients define the street. **T:** maximise the margin $2/\|w\|$ subject to $y_i(w^\top x_i - b) \geq 1$; the dual depends on data only through inner products $x_i^\top x_j$; slack variables allow soft margins (hyper-parameter C). **H:** widely used for disease classification [P6]; [P7]'s tuned RBF-SVM is the model to beat (AUC 0.998 internal). **W:** it is (a) a strong classical baseline and (b) the classical object the QSVM generalises — the QSVM is “SVM with a quantum kernel.” **M:** not scaling before an SVM; not tuning C and γ . **P:** block C2 (classical) and Q3a (quantum kernel).

RBF kernel. **S:** similarity that decays with distance: near points very similar, far points not at all. **T:** $K_{\text{RBF}}(x_i, x_j) = \exp(-\gamma\|x_i - x_j\|^2)$; corresponds to an infinite-dimensional feature space; γ sets the “reach.” **H:** [P6] uses RBF as the classical kernel benchmark; GEKO kernels “match or beat” it. **W:** the RBF-SVM is the fair classical opponent for our QSVM; the external “inductive bias of quantum kernels” result (Kübler 2021) says a *bandwidth-tuned* RBF often dominates quantum kernels — so we must tune γ . **M:** comparing a tuned quantum kernel to an *untuned* RBF. **P:** block C2.

Hyper-parameter tuning. **S:** the knobs you set before training (e.g. C , γ , tree depth, #layers). **T:** grid / random / Bayesian (Optuna TPE, [P7]) search, scored by CV; budget must be *equal* across classical and quantum models ([P7]'s explicit fairness rule; [P7]'s QSVM was *not* equally tuned and it shows). **H:** [P4] used GridSearchCV; [P7] used Optuna. **W:** an unfair tuning budget is the #1 way QML papers manufacture a fake win [P5]. **M:** tuning the quantum model hard and the classical model not at all (or vice-versa). **P:** block C2–C4 and Q2–Q3.

Random Forest / gradient boosting (XGBoost, LightGBM, CatBoost). **S:** many decision trees voting; boosting builds trees that fix the previous trees' mistakes. **T:** bagging (RF) vs. additive stagewise boosting; strong on tabular data, little scaling needed. **H:** [P7]: XGBoost/LightGBM/CatBoost all AUC 0.99+ on WDBC; [P5]: these are the baselines QML papers should use but often don't. **W:** these are the true state of the art on health *tables* — our headline classical opponents. **M:** using default hyper-parameters; [P7] notes LightGBM fails degenerately below ~ 20 training rows. **P:** blocks C3–C4.

7.6 Evaluation metrics

Metric	Definition	Use / caution (papers)
Accuracy	$(\text{TP} + \text{TN}) / \text{all}$	Misleading under imbalance [P5]; never the headline.

Precision	$TP/(TP+FP)$	“Of those we flagged, how many were sick.”
Recall / Sensitivity	$TP/(TP+FN)$	“Of the sick, how many we caught” — the priority in screening [P4][P5].
Specificity	$TN/(TN+FP)$	“Of the healthy, how many we cleared.”
F1	harmonic mean of P & R	Balances P/R; ignores TN [P5].
ROC-AUC	rank-quality across all thresholds	Standard [P4][P7]; over-optimistic under heavy imbalance [P5].
PR-AUC	precision–recall area	<i>Preferred</i> under imbalance [P5]; our headline metric.
MCC	correlation of prediction & truth	Robust single number [P7].
Brier score	mean squared error of probabilities	Calibration + resolution [P7].
Reliability diagram / ECE	predicted vs. observed frequency	Is “0.8” really 80%? [P7].
Decision-curve analysis	net benefit vs. threshold	Translates AUC into clinical value [P7].
DeLong / bootstrap CI	significance of an AUC gap	Stops you over-reading a 0.5% difference [P7][P10].

KEY IDEA

For early detection our reporting bundle is: **PR-AUC** (headline) + ROC-AUC + recall-at-fixed-precision + Brier + reliability diagram + decision-curve analysis + 95% CIs + DeLong test for every quantum-vs-classical gap. This is exactly [P7]’s bundle, plus PR-AUC for imbalance.

8 Quantum basics (from absolute zero, only what we need)

Format per concept: **intuition** → **technical** → **tiny example** → **relevance to our system**.

8.1 Bit vs. qubit

Intuition. A bit is a switch: 0 or 1. A qubit is like a spinning arrow that can point anywhere on a globe; only when you “look” (measure) does it snap to 0 or 1, with a probability set by where it was pointing. **Technical.** A qubit state is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $\alpha, \beta \in \mathbb{C}$, $|\alpha|^2 + |\beta|^2 = 1$. **Example.** $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ gives 0 or 1 with probability $\frac{1}{2}$ each. **Relevance.** Each PCA component of a patient becomes a rotation angle that sets where one qubit’s arrow points.

8.2 Superposition

Intuition. “Both at once, in a specific blend” — not “we don’t know which.” **Technical.** any $\alpha|0\rangle + \beta|1\rangle$ with both non-zero. n qubits live in a superposition over all 2^n bitstrings. **Example.** 8 qubits $\Rightarrow$ a vector of 256 complex amplitudes. **Relevance.** This 2^n -dimensional space is the “quantum feature space” our feature map writes into (Part 10).

8.3 Measurement, probability amplitudes, expectation values, shots

Intuition. You can’t read the arrow directly; you get a coin flip whose bias depends on the arrow. Flip many times to estimate the bias. **Technical.** outcome k has probability $|\langle k|\psi\rangle|^2$. For observable $\hat{O}$, the *expectation value* is $\langle\hat{O}\rangle = \langle\psi|\hat{O}|\psi\rangle$; estimated from M repetitions (“shots”) with error $\sim 1/\sqrt{M}$. **Example.** $\langle Z\rangle = P(0) - P(1)$; with 1000 shots the estimate is good to ~ 0.03 . **Relevance.** A VQC prediction *is* an estimated $\langle Z\rangle$; a quantum kernel entry *is* an estimated overlap probability. Finite shots inject noise — part of our uncertainty budget (block 10).

8.4 Quantum state & Bloch-sphere intuition

Intuition. One qubit = a point on a globe (Bloch sphere): north pole $|0\rangle$, south pole $|1\rangle$, equator = equal superpositions differing by “phase.” **Technical.** $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$; (θ, φ) are the globe coordinates. **Example.** $\theta = \pi/2, \varphi = 0$ is $|+\rangle$. **Relevance.** [P2] literally trains rotations that move the patient’s state to the Bloch-sphere region of its class; [P7]’s Z-rotation failure is a Bloch-sphere fact (rotating about the measurement axis does nothing visible).

8.5 Quantum gates

Intuition. Gates are rotations of the arrow(s). **Technical.** unitary matrices U ($U^\dagger U = I$), reversible.

X, Y, Z (Pauli)

180° flips about the $x/y/z$ axes. $X = \text{NOT}$; $Z = \text{phase flip}$.

Hadamard H

maps $|0\rangle \rightarrow |+\rangle$ — creates an equal superposition. The “open the door to quantumness” gate.

Rotation gates $R_x(\theta), R_y(\theta), R_z(\theta)$

rotate by a *continuous* angle θ about an axis. These carry both our *data* (angle = feature value) and our *trainable parameters* (angle = θ_i).

CNOT (CX)

two-qubit: flip the target iff the control is $|1\rangle$. Creates *entanglement*. CZ is the phase cousin used in [P2].

Relevance. Our circuits are exactly: $H / R_x / R_y / R_z / \text{CNOT}$ — the same alphabet as [P2][P4][P6][P7].

8.6 Entanglement

Intuition. Two qubits so linked that measuring one instantly tells you about the other — the pair has a joint description that can’t be split into “qubit A’s state” and “qubit B’s state.” **Technical.** a state that is not a tensor product, e.g. $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. **Example.** apply H then CNOT to $|00\rangle$.

Relevance. Entangling layers let the feature map/ansatz build *correlations between features* — [P2] shows entanglement improves multi-qubit data-efficiency; [P6] finds kernel entanglement entropy is *weakly* correlated with accuracy (more is not automatically better).

8.7 Quantum circuits & parameterised (variational) circuits

Intuition. A circuit is a score: qubit “wires” left-to-right with gate “notes.” A *parameterised* circuit leaves some gate angles as adjustable dials. **Technical.** $U(\vec{\theta}) = U_L(\theta_L) \cdots U_1(\theta_1)$; a PQC/VQC/ansatz. **Example.** $R_y(\theta_1)$ on q0, $R_y(\theta_2)$ on q1, CNOT(0,1), measure Z_0 . **Relevance.** Our VQC arm *is* a PQC; our QSVM arm uses a PQC as the (possibly optimised) feature map.

8.8 Noise & NISQ

Intuition. Real qubits are leaky: gates are slightly wrong, qubits forget over time (T_1/T_2), readout misfires, neighbours interfere (crosstalk). **Technical.** modelled as channels — depolarising, amplitude damping, phase damping, readout confusion matrix. NISQ = this regime, no error correction [P1][P4][P5]. **Example.** [P7]: depolarising $p = 0.1$ per qubit barely dented its VQC — but it *omitted* T_1/T_2 , coherent and correlated noise, which “can dominate on real devices.” **Relevance.** Block 9’s noise sweep; our honesty about what our noise model does and does not include.

8.9 Quantum simulators vs. real hardware

Intuition. A simulator runs the linear algebra on a normal computer (exact up to ~ 30 – 40 qubits); real hardware is a physical device you queue for. **Technical.** statevector sim (exact, noiseless), density-matrix / stabiliser / tensor-network sims (noisy, larger), “noisy simulator” = statevector + channels. **Example.** [P7] used PennyLane `lightning.qubit + default.mixed`; *no physical QPU*. [P10]: 131/138 sim-only health studies used *ideal* sims. **Relevance.** We develop on statevector + noisy simulators; budget for *one* short real-QPU run of the final kernel (free-tier IBM / IonQ) purely as a reality check, clearly labelled.

THE 6 QUANTUM FACTS YOU MUST BE ABLE TO SAY IN ONE BREATH

(1) n qubits = a 2^n -dim complex vector. (2) Gates are reversible rotations; measurement is a biased coin. (3) You estimate everything from finite *shots*, so there is always sampling noise. (4) Data enters as *rotation angles*; training parameters are *also* rotation angles. (5) Entanglement builds feature correlations. (6) We are in the *NISQ* era — few qubits, real noise, no error correction — so circuits must be **shallow and narrow**.

9 Data encoding — classical numbers into a quantum state

9.1 Why classical data must be encoded

A quantum circuit only acts on quantum states. So a patient’s feature vector x must first be *written into* a state $|\phi(x)\rangle$. This step is not free: it costs qubits and circuit depth, and its choice strongly affects both accuracy and trainability [P2][P3][P10]. [P10] (Table 1) summarises the cost trade-offs:

Encoding	Qubits	Depth / run-time	Idea & notes
Basis	$O(d\tau)$	$O(1)$ per vector	Put the bits of x directly on qubits. Exact, wasteful in qubits.
Amplitude	$O(\log_2(d\tau))$	$O(Nd\tau)$ general	Pack d numbers into the 2^n amplitudes. Few qubits, but <i>linear-time</i> state prep in general kills speed-ups [P10]; used by [P8].
Angle / rotation	$O(d)$	$O(1)$	One feature $\rightarrow$ one rotation angle on one qubit. Simple, shallow, hardware-friendly. Used by [P4] (<code>AngleEmbedding</code> , R_x), [P7] (R_y).
Data re-uploading	$O(d/3)$ qubits	$O(L)$ layers	Feed the features in <i>repeatedly</i> , between trainable layers — raises expressivity with few qubits [P2].
QRAM / parallel loaders	sublinear	favourable	Assume special hardware that does not exist at scale [P10]. Out of scope.

9.2 The four encodings we care about

Basis encoding. $x = 101 \Rightarrow |101\rangle$. Only for already-binary features; we won’t use it.

Amplitude encoding. normalise x to unit length, then $|\phi(x)\rangle = \sum_k x_k |k\rangle$ over $\lceil \log_2 d \rceil$ qubits. Pro: 8 features in 3 qubits. Con: the state-preparation circuit is deep and generally $O(d)$ — [P10] and [P3] both warn this cost negates asymptotic advantage. [P8] uses it on PCA’d genomic data; we treat it as an *advanced ablation*, not the default.

Angle encoding. $x_j \mapsto R_y(x_j)$ (or R_x, R_z) on qubit j . Depth 1, d qubits, hardware-native. *This is our default.* Scale features to a safe angle range first ($[0, \pi]$ [P7], or $[-\pi/2, \pi/2]$ [P6]).

Rotation / Pauli feature maps. angle encoding *plus* entangling rotations $e^{i\phi_S(x)} \prod_{j \in S} P_j$ (the ZZ / Pauli-expansion map, Havlíček 2019) — adds feature *products* as phases. More expressive; often a poor inductive bias [P6][P7]. This is the *baseline* map our search tries to beat.

THE Z-ROTATION TRAP (from [P7])

If you angle-encode with R_z acting on $|0\rangle$ and then measure $\langle Z \rangle$, the encoding produces only a global phase — **the data becomes invisible** and the model collapses to chance ([P7]: AUC ≈ 0.50 at every depth). Rule: the encoding rotation axis must *not commute* with the read-out observable. Use R_y or R_x with a Z read-out. This is a mandatory ablation in our plan.

9.3 Worked example: a five-feature patient

Raw record: Age = 58 yr, BMI = 31.4, Fasting glucose = 142 mg/dL, Systolic BP = 150 mmHg, Total cholesterol = 236 mg/dL. Label: at risk (1).

STEP 1	RAW	Age 58 BMI 31.4 Glu 142 SBP 150 Chol 236
STEP 2	CLEAN	no missing; units checked; ranges plausible
STEP 3	SCALE	z-score using TRAIN-FOLD mean/SD -> (fit on train fold only!) Age 0.9 BMI 1.1 Glu 1.6 SBP 1.2 Chol 0.7
STEP 4	SELECT/PCA	5 features -> PCA (train-fold fit) -> 4 components ($\geq 95\%$ var) z = [1.42, -0.83, 0.19, 0.55]
STEP 5	ANGLE MAP	scale each comp to $[0, \pi]$: a = [2.6, 0.4, 1.7, 2.1]
STEP 6	ENCODE	q0: Ry(2.6) q1: Ry(0.4) q2: Ry(1.7) q3: Ry(2.1) (baseline ZZ map adds: CNOT ring + Rz((pi - a_i)(pi - a_j)) on pairs)
STEP 7	CIRCUIT	q0: --Ry(2.6)--*-----Rz(.)--- ... (feature-map body) ... --[measure] q1: --Ry(0.4)--X-----Rz(.)--- q2: --Ry(1.7)-----X-----Rz(.)--- q3: --Ry(2.1)-----X-----Rz(.)---
STEP 8	USE	QSVM: run this + its adjoint for pairs -> kernel entry VQC : append trainable layers -> measure $\langle Z_0 \rangle$ -> probability

9.4 Dimensionality constraints & the PCA–qubit link

- #qubits in angle encoding = #features fed in. NISQ budget $\Rightarrow$ 4–8.
- So: select ≤ 8 informative features (block 5), or PCA to $d \leq 8$ components (block 6). [P6] used “#components for 95% variance”; [P7] used “8 = qubit budget.”
- Guard against post-PCA collapse: check that no two *different-label* training points map to (near-)identical encoded vectors ([P10]’s duplication warning). Report this check.

IMPLEMENTATION NOTE

Default pipeline: RobustScaler $\rightarrow$ SelectKBest(mutual_info, k=8) *or* PCA(n_components=d) $\rightarrow$ min-max to $[0, \pi]$ $\rightarrow$ AngleEmbedding(rotation="Y"). All fitted inside the CV fold. Amplitude encoding and $d \in \{4, 6, 8\}$ sweeps are ablation arms.

10 Quantum feature maps

10.1 Build-up: what is a feature map in ordinary ML?

Step 1 — the problem. Some data can't be separated by a straight line. Example: points inside a circle are class 1, outside are class 0. No line works.

Step 2 — the fix. Send each point through a function ϕ into a *higher* space where a plane *does* separate them. For the circle, $\phi(x_1, x_2) = (x_1, x_2, x_1^2 + x_2^2)$ adds a “height” axis; now a horizontal plane cuts inside from outside.

Step 3 — the kernel shortcut. SVMs only ever need *inner products* $\phi(x_i)^\top \phi(x_j)$, never ϕ itself. So define a *kernel* $K(x_i, x_j) = \phi(x_i)^\top \phi(x_j)$ and skip building the big vectors — the “kernel trick” [P6].

10.2 What changes when the feature space is quantum?

The feature map is now a *circuit* $U(x)$, and the “big vector” is a quantum state:

$$|\phi(x)\rangle = U(x)|0\rangle^{\otimes n} \in \mathbb{C}^{2^n}.$$

The target space has dimension 2^n — exponentially large in the number of qubits. The induced kernel is the squared overlap of two states:

$$K(x_i, x_j) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2 = |\langle 0 | U^\dagger(x_i) U(x_j) | 0 \rangle|^2.$$

Intuition first: run the circuit that prepares $|\phi(x_j)\rangle$, then run the *reverse* of the circuit for $|\phi(x_i)\rangle$; measure the probability of landing back in $|0\dots 0\rangle$. If the two encoded patients are “similar” that probability is high. That single number is one entry of the kernel matrix [P6][P9].

Classical feature map	Quantum feature map
-----	-----
x --phi--> phi(x) in R ^D	x --U(x)--> phi(x)> in C ^(2ⁿ)
K = phi(x _i).phi(x _j)	K = <phi(x _i) phi(x _j)> ^2
D chosen by hand (poly)	2 ⁿ (exponential, but mostly unused)
overlap circuit for one kernel entry:	
0...0> --[U(x _j)]--[U(x _i) [†]]--[measure all]--> P(0...0) = K(x _i , x _j)	

10.3 Ingredients of a quantum feature map

$U(x)$

the whole data-encoding unitary.

Parameterised encoding

angles are functions of x (linear: $R_y(x_j)$; or products: $R_z((\pi - x_i)(\pi - x_j))$ as in the ZZ map).

Entangling feature maps

insert CNOT/CZ so the map encodes *feature interactions*, not just marginals [P2][P6].

Repetitions / re-uploading

apply the encoding block r times ([P7] used ZZ “reps= 2”; [P2] re-uploads L times) — more reps = more expressive, deeper, harder to train.

10.4 Expressivity vs. trainability — the central trade-off

KEY IDEA

A more expressive map reaches more of the 2^n -dimensional space. But [P3] shows that when the map spreads states *too* thinly, *all* pairwise overlaps become tiny and similar — the kernel matrix $\rightarrow$ identity, the “geometric difference” from a classical kernel shrinks, and the model **loses**, not gains. Likewise [P1]: highly expressive (near-random) circuits have exponentially vanishing gradients (barren plateaus). **Sweet spot: expressive enough to capture the data’s structure, shallow/narrow enough to train and to keep overlaps meaningful.**

10.5 Feature-map selection

Options, cheapest first:

1. **Fixed families with a few knobs** — angle map; ZZ map with reps $\in \{1, 2, 3\}$; entanglement pattern $\in \{\text{linear, circular, full}\}$. Grid-search inside CV. (This alone is more than most papers do.)
2. **Kernel-target alignment (our MVP optimiser)** — put trainable angles λ in the map, maximise the alignment between K_λ and the ideal label kernel $yy^\top$: $\hat{A}(K_\lambda, yy^\top) = \frac{\langle K_\lambda, yy^\top \rangle_F}{\|K_\lambda\|_F \|yy^\top\|_F}$. Gradient ascent on λ , all inside the training fold (external: Hubregtsen et al. 2022).
3. **Genetic / evolutionary search (advanced — GEKO, [P6])** — population of gate sequences from $\{X, \sqrt{X}, R_z, \text{CNOT}\}$; fitness = inner-CV accuracy (supervised) or max normalised kernel eigenvalue (unsupervised); mutate, select, iterate; binary-search the gene count for the smallest circuit that hits target fitness.

10.6 Hardware-aware feature maps

Restrict the gate pool to the target device’s *native* gates and respect its qubit *connectivity* ([P6] used IBM-native $\{X, \sqrt{X}, R_z, \text{CNOT}\}$). This avoids compilation blow-up ([P10]: transpiling a non-native 2-qubit gate can multiply gate count).

RESEARCH GAP

[P6] shows genetic search *beats* the fixed ZZ map on 9 datasets — but on a single split family, ideal simulator, no external cohort, and *no report of whether the search overfit the validation signal*. Our contribution is to run map selection under nested CV, validate the winner on an external cohort, and publish the search’s compute budget and an overfitting check.

11 QSVM — the deepest section

11.1 Start from classical SVM (intuition before maths)

Two clouds of patients, sick and healthy. Draw a line between them — but not just any line: the one with the widest empty “street” on both sides. The patients touching the kerb are the *support vectors*; they alone define the line. A wide street means small changes in a new patient won’t flip the prediction — the classifier is *robust*.

11.2 The pieces

Decision boundary

the surface where $P(\text{sick}) = P(\text{healthy})$; for a linear SVM, $w^\top x - b = 0$.

Margin

the width of the street, $2/\|w\|$. SVM *maximises* it.

Support vectors

the training points on the margin edges (or inside it for soft margins).

Soft margin (C)

real data overlaps; C trades street width against violations. Small C = wide street, more mistakes tolerated; large C = narrow street, fit training data hard.

11.3 Nonlinear classification & the kernel trick

When no line works in the original space, map to a richer space $\phi(x)$ where one does. The SVM dual only uses inner products, so replace $\phi(x_i)^\top \phi(x_j)$ by a *kernel* $K(x_i, x_j)$ and never build ϕ explicitly:

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i y_i = 0.$$

Classical kernels: linear $x_i^\top x_j$; polynomial $(x_i^\top x_j + c)^p$; RBF $\exp(-\gamma \|x_i - x_j\|^2)$ [P6].

11.4 From classical kernel to quantum kernel

Intuition. Instead of a formula for similarity, *prepare* each patient as a quantum state and measure how much the two states overlap.

Technical [P6][P9]. Encode x with a feature-map circuit: $|\phi(x)\rangle = U(x)|0\rangle^{\otimes n}$. Define the quantum kernel as the squared state overlap (a.k.a. fidelity):

$$K(x_i, x_j) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2 \quad \text{with} \quad 0 \leq K \leq 1, \quad K(x, x) = 1.$$

11.5 Estimating one kernel entry

Run the *overlap* (or “compute–uncompute”) circuit and count how often you measure all zeros:

$$K(x_i, x_j) = |\langle 0 | U^\dagger(x_i) U(x_j) | 0 \rangle|^2 = \lim_{M \rightarrow \infty} \frac{\#(\text{outcome } 00 \dots 0)}{M}.$$

```

|0...0> --[ U(x_j) ]--[ U(x_i)^\dagger ]--[ measure Z^n ]--> count 00...0 / M
repeat for every pair (i,j) -> N x N kernel matrix K

```

Cost: $O(N^2)$ circuit families, each run M times. [P7] calls this “ $O(N^2)$ scaling ... made an iterative Bayesian optimisation loop computationally prohibitive” — so it left the QSVM *untuned*, and it underperformed. *We will budget for this explicitly* (subsample N , report it).

11.6 Training & prediction

Train: build $K_{\text{train}} (N \times N)$, hand it to a classical SVM solver (convex QP) $\rightarrow$ get α_i, b . **Predict** a new patient x_* : compute $K(x_*, x_i)$ for each support vector, then $\hat{y}(x_*) = \text{sign}(\sum_i \alpha_i y_i K(x_*, x_i) + b)$; a probability comes from Platt scaling.

11.7 The quantum / classical split

Quantum device	estimate kernel entries $K(x_i, x_j)$ (only this)
Classical computer	solve the SVM QP, choose C , Platt-scale, predict, evaluate

11.8 Limitations (say them plainly)

- **Cost:** $O(N^2M)$ circuit runs; impractical for large N [P4][P7].
- **Inductive bias:** the fixed ZZ map is often mis-aligned with the data geometry — [P7] AUC 0.727; [P6] ZZ worst on all 9 datasets. (This is what our search targets.)
- **Kernel concentration / exponential dimension:** at many qubits or a too-expressive map, off-diagonal $K \rightarrow 0$, $K \rightarrow I$, generalisation collapses [P3] (external: Thanasilp et al. 2024, “Exponential concentration in quantum kernel methods,” *Nat. Commun.*, arXiv:2208.11060).
- **Noise:** shot noise makes K non-PSD (fixable by regularisation); hardware noise flattens K [P1].
- **Dequantisation:** for small classical datasets, a sufficiently expressive classical kernel can match a quantum kernel [P3] (external: Kübler et al. 2021).

11.9 Complete tiny healthcare workflow

```

DATA    6 patients, 2 PCA features each, label sick/healthy
        p1 (0.9,0.2) S    p2 (0.8,0.1) S    p3 (0.2,0.9) H
        p4 (0.1,0.8) H    p5 (0.85,0.15) S p6 (0.15,0.85) H

ENCODE  U(x): q0<-Ry(pi*x1), q1<-Ry(pi*x2), CNOT(0,1), Rz((pi-x1)(pi-x2)) on q1

KERNEL  K(p1,p2) = |<phi(p1)|phi(p2)>|^2 ~ 0.98    (both "sick" corner)
        K(p1,p3) ~ 0.05                            (opposite corners)
        ... fill 6x6 matrix ...

TRAIN   classical SVM on K -> support vectors {p2,p3,p5,p6}, C=1

PREDICT new patient x* = (0.8, 0.25):
        k = [K(x*,p2), K(x*,p3), K(x*,p5), K(x*,p6)] ~ [0.95, 0.06, 0.97, 0.05]
        score = sum a_i y_i k_i + b = +1.7 -> SICK, Platt prob 0.91

EVALUATE do this inside CV; compare to RBF-SVM on the SAME folds; DeLong test

```

KEY IDEA

QSVM = ordinary SVM + a kernel whose entries are measured state overlaps. The only quantum part is the kernel. Our project’s lever: *stop hand-picking the feature map that defines that kernel; select it from data, leakage-free, and check it honestly against a tuned RBF-SVM.*

12 QNN / VQC — the trainable quantum circuit

12.1 One-sentence idea

A QNN/VQC is a small quantum circuit with adjustable gate angles, trained like a neural network: run it, measure a number, compare to the label, nudge the angles, repeat [P2][P7][P9].

12.2 The parts

Encoding

write x into the state (angle encoding, Part 9).

Ansatz / variational circuit

alternating layers of trainable single-qubit rotations $R(\theta)$ and fixed entanglers (CNOT/CZ ring). [P7]: 4 “StronglyEntanglingLayers”; [P4]: 4 “BasicEntanglerLayers.”

Trainable parameters θ

the rotation angles inside the ansatz. [P7]: $4 \times 8 \times 3 = 96$.

Quantum layers

one encode+ansatz block; stack L of them (with data re-uploading, Part 13).

Measurement

read an observable, usually $\langle Z_0 \rangle$ on the first qubit.

Expectation value $\rightarrow$ prediction

rescale $\langle Z_0 \rangle \in [-1, 1]$ to a probability $p \in [0, 1]$ ([P7]: $(\langle Z_0 \rangle + 1)/2$).

Loss

binary cross-entropy ([P7]) or a fidelity cost ([P2]).

Classical optimiser

Adam ([P7]), L-BFGS-B ([P2]), or SPSA (shot-frugal).

Gradient / parameter-shift rule

the exact gradient of an expectation w.r.t. a rotation angle: $\frac{\partial \langle O \rangle}{\partial \theta_k} = \frac{1}{2} [\langle O \rangle_{\theta_k + \pi/2} - \langle O \rangle_{\theta_k - \pi/2}]$
 — evaluate the *same circuit* at two shifted angles [P6].

Training loop

see diagram.

Data re-uploading

re-inject x between layers to boost expressivity with few qubits (Part 13).

Overfitting / trainability

few qubits + shallow = less overfitting but watch barren plateaus (Part 14).

12.3 The training loop

```

      x  (one patient, d PCA features)
      |
      v
[ ENCODE  U(x) ]      angle-encode d features on d qubits
      |
      v
[ ANSATZ  V(theta) ]  L layers: Rx/Ry/Rz(theta_i) + CNOT ring
      |
      v
[ MEASURE <Z_0> ]      ~ estimate from M shots
      |
      v
p = (<Z_0> + 1) / 2  ->  prediction
      |
      v
loss = BCE(p, y)
      |
      v
grad via PARAMETER-SHIFT: d<Z_0>/dtheta_k = (1/2) [ <Z_0>_{+pi/2} - <Z_0>_{-pi/2} ]
      |
      v
theta <- theta - eta * grad      (Adam / SPSA)
      |
      +----- repeat over mini-batches / epochs until val loss stops improving

```

12.4 Prediction

After training: encode the new patient, run $V(\theta^*)$, measure $\langle Z_0 \rangle$, rescale to p , threshold at a chosen operating point (Part 16), attach an uncertainty interval (Part 17).

12.5 Overfitting & trainability notes

- Few qubits + shallow ansatz $\Rightarrow$ limited capacity $\Rightarrow$ overfitting is less of a worry than in deep nets — but *do* use a validation-loss early stop ([P4]: patience 10) and report a learning curve

([P7]).

- [P2] warns of *local minima* in the (trigonometric) loss landscape — use several random seeds, report spread.
- Barren plateaus (Part 14) bite when the ansatz is deep/wide/expressive — our design rules keep us out of that regime.

SIH JUDGE NOTE

“Is your VQC a neural network?” — *Functionally yes*: a parameterised function trained by gradient descent on a loss. *Structurally no*: the “neurons” are qubit rotations, the nonlinearity comes from measurement and from re-uploading, and the number of effective parameters is tiny (tens, not millions). This is why it is a candidate for *small-data* regimes — and why it will never out-muscle a deep net on big data.

13 Data re-uploading (from [P2])

13.1 What it is

Instead of encoding the data once at the start, feed it in *again and again* between trainable layers:

$$U(\vec{\phi}, \vec{x}) = \underbrace{U(\vec{\phi}_L)U(\vec{x})}_{L(L)} \cdots \underbrace{U(\vec{\phi}_1)U(\vec{x})}_{L(1)}.$$

Each “processing layer” $L(i) = U(\vec{\phi}_i)U(\vec{x})$ is a data upload followed by a trainable rotation. [P2] also compresses both into one gate: $L(i) = U(\vec{\theta}_i + \vec{w}_i \circ \vec{x})$ where $\vec{w}_i \circ \vec{x}$ is an element-wise product — the $\vec{w}_i$ act like neural-network weights, $\vec{\theta}_i$ like biases.

13.2 Why it was proposed

A single qubit has only two degrees of freedom and one rotation cannot separate non-trivial patterns. The *no-cloning theorem* forbids copying quantum data — but a classical feature vector *can* be re-fed. [P2] shows this recovers the power a neural network gets from processing its input through many hidden neurons: it proves an equivalence with the *Universal Approximation Theorem* — a single qubit with enough re-uploads is, in principle, a universal classifier.

13.3 Architecture

```
|0> --[U(x)]--[U(phi_1)]--[U(x)]--[U(phi_2)]-- ... --[U(x)]--[U(phi_N)]--[measure]
      \_____ layer 1 _____/ \_____ layer 2 _____/          layer N

compressed: |0> --[U(theta_1 + w_1 o x)]-- ... --[U(theta_N + w_N o x)]--[measure]

multi-qubit + entanglement:
q0 --L1(1)---*---L1(2)---*--- ... --L1(N)--
q1 --L2(1)---X--L2(2)---X-- ... --L2(N)--    (CZ/CNOT between layers)
```

13.4 Repeated encoding & trainable layers

- **More layers** $N \Rightarrow$ more expressive (like more hidden neurons). [P2] circle task: 1 layer only splits the plane in half; 2 layers catch the circle; more layers refine the radius.
- **Parameter count** (single qubit): $(3 + d)N$ (d = input dimension) + C class weights if using the weighted-fidelity cost [P2].
- **Cost split into the classical part**: [P2] deliberately grows the classical optimisation to shrink the quantum resource (fewer layers) — the right instinct for NISQ.

13.5 One-qubit intuition → multi-qubit

- **One qubit:** each layer is a Bloch-sphere rotation; many different rotations together build an effective feature map; data in that feature space is separated by regions of the sphere [P2].
- **Multi-qubit + entanglement:** acts like adding hidden *layers* (depth), not just width — fewer layers reach the same accuracy, and [P2] argues entanglement is where a genuine quantum advantage could enter. On the 4-D hypersphere task, 4 qubits + entanglement hit 98% with just 2 layers vs. 8 layers for one qubit.

13.6 Advantages / limitations

Advantages: minimal qubits; shallow per-layer; naturally nonlinear; hardware-friendly; strong on structured low-dimensional data.

Limitations: [P2]’s benchmarks are *synthetic / tiny* (circle, annulus, hypersphere) with no health validation; local minima; success rate can *drop* when adding layers (optimiser gets trapped); no external validation, no noise study in the source.

HOW WE USE IT

Our **VQC arm (block Q3b)** is a data-re-uploading circuit: d qubits ($d \in \{4, 6, 8\}$), $L \in \{2, 3, 4\}$ re-uploading layers, per-layer $R_x R_y R_z$ + CNOT ring, $\langle Z_0 \rangle$ read-out, BCE loss, Adam, parameter-shift gradients, early stop on validation loss, 10 seeds. L is a hyper-parameter tuned in the inner CV with the *same budget* as the classical models. We do *not* attempt the universality proof — we use re-uploading purely as a practical expressivity booster for few qubits.

14 Barren plateaus (from [P1])

14.1 The concepts

Gradient

the slope of the loss w.r.t. each parameter — the direction the optimiser steps.

Optimisation landscape

loss as a surface over parameter space; training walks downhill on it.

Barren plateau

a vast, almost perfectly flat region of that surface — the gradient is essentially zero *everywhere*, so the optimiser has no direction to go and training stalls.

Why gradients vanish [P1]

for a wide class of *random* parameterised circuits, as the qubit count n grows, the expected gradient is 0 and its variance shrinks *exponentially in n* : $\text{Var}[\partial_k E] \sim 2^{-\alpha n}$. Geometrically: “concentration of measure” — in high dimension, almost all states look average, so almost all observable values (and their derivatives) sit at the mean.

Relationship to circuit depth

you don’t even need a deep circuit — [P1] shows $O(n^{1/d})$ depth on a d -dimensional qubit lattice is enough for the circuit to behave like a random 2-design and land you on the plateau.

Random circuits

the culprit: circuits initialised with random angles and enough expressivity to approximate the Haar (uniform) distribution.

Noise

makes it worse: *noise-induced* barren plateaus arise even for shallow circuits as noise drives the state toward the maximally-mixed state (external: Wang et al., “Noise-induced barren plateaus in variational quantum algorithms,” *Nat. Commun.* 12:6961, 2021).

Practical consequence

estimating a gradient on hardware costs $O(1/\varepsilon^\alpha)$ shots; if the true gradient is exponentially small, you'd need exponentially many shots to even see it — so gradient descent becomes a random walk with exponentially small chance of escaping [P1].

14.2 Contrast with classical deep nets

Classical vanishing gradients scale with the number of *layers* and can often be fixed (normalisation, residual connections, careful init). Quantum barren plateaus scale with the number of *qubits*, are generally *smaller*, and the measurement cost to fight them is *worse* [P1].

14.3 What this means for OUR QNN — practical design rules

BARREN-PLATEAU DESIGN RULES FOR OUR CIRCUITS

1. **Stay small:** $d \leq 8$ qubits. At this width, plateaus are usually avoidable empirically ([P7] trained an 8-qubit VQC fine).
2. **Stay shallow:** $L \leq 4$ layers; avoid “hardware-efficient” ansatz stacked deep.
3. **Don't use fully random init:** use small-angle initialisation ($\theta \sim \mathcal{N}(0, 0.1)$) or an identity-block / layerwise-warm-start init.
4. **Prefer structured / problem-inspired ansatz:** data re-uploading [P2], or a local-cost ansatz; avoid global-cost observables (measure $\langle Z_0 \rangle$, not a projector onto $|0 \dots 0\rangle$).
5. **Use a local cost function:** [P1]'s worst case is the global projector cost; a single-qubit Z read-out is far safer.
6. **Measure the landscape:** *report the gradient-variance vs. layer count / qubit count for our exact circuit* — [P10] notes **no** health-QML study does this. It is ~ 20 lines of PennyLane and is a credibility win.
7. **Shot budget:** use SPSA or a moderate shot count; log how many circuit evaluations training actually cost ([P5][P10] ask for this).
8. **Watch noise:** include the noise sweep (Part 17); if gradients die under noise, say so.

RECENT LITERATURE / EXTERNAL EVIDENCE

Recent theory (Larocca et al., “A review of barren plateaus in variational quantum computing,” arXiv:2405.00781, 2024; Cerezo et al., “Does provable absence of barren plateaus imply classical simulability?,” arXiv:2312.09121) suggests that circuits provably *free* of barren plateaus may also be *classically simulable* — i.e. the trainability you need and the quantum hardness you want may be in tension. For an SIH prototype this is a talking point, not a blocker: we operate in the small-scale regime where we can train, we simulate anyway, and we are honest that scale-up trainability is an open research problem.

15 Quantum advantage — and how we would actually test for it (from [P3])

15.1 The vocabulary, carefully

Quantum advantage

an *asymptotic* reduction in resources (queries, samples, time) as the problem size $\rightarrow \infty$ [P10].
A mathematical statement about scaling.

Quantum speed-up

specifically a reduction in *time* complexity [P10].

Empirical quantum advantage / utility

a finite-size experiment where quantum beats classical on a task, *assuming* the gap will persist at scale [P10]. Weaker; easy to fake.

Quantum utility

loosely, “the quantum model is useful here” — [P10] calls this term “poorly defined and frequently abused.”

Dequantisation

a classical algorithm is found that matches the quantum one — the advantage evaporates [P3][P10].

15.2 Why accuracy alone is not evidence of advantage

[P3]’s central result: **with training data, classical ML can match quantum models — even on data generated by quantum circuits that are hard to simulate.** A quantum function $f(x) = \langle x|U^\dagger O U|x\rangle$ can be hard to *compute* from scratch (Proposition 1: if you could, then $\text{BPP} = \text{BQP}$) yet easy to *learn* from examples, because it is a quadratic form with p^2 coefficients and a classical model achieves error $\lesssim \sqrt{p^2/N}$. So “our quantum model got 92%” says nothing about advantage — a classical model given the same data may get 92% too.

15.3 The role of data, representation, kernel choice, baseline

- **Data:** more training data lifts the classical model, shrinking any gap [P3].
- **Representation / kernel:** advantage lives in the *geometry* of the kernel, not in “using a quantum computer.” If the quantum and classical kernels induce similar geometries, classical wins or ties [P3].
- **Baseline:** must be the *best* tuned classical model, chosen from a suite [P3][P7] (external: Bowles et al. 2024).
- **Scalability:** more qubits can *hurt* — the naive quantum kernel’s Hilbert space grows so fast that all inner products become tiny, the effective dimension $d \propto N$, and the geometric difference collapses [P3].

15.4 [P3]’s testing methodology (the tools we adopt)

INPUT: N samples, a classical kernel K_C, a quantum kernel K_Q (or projected K_PQ)

- (1) GEOMETRY TEST g_CQ -- function-INDEPENDENT pre-screen
 g_CQ small -> classical ML guaranteed to match/beat quantum. STOP.
 g_CQ large -> a dataset with a quantum prediction advantage EXISTS.
 - (2) DIMENSION TEST d (effective dim of the quantum feature space)
 d ~ N -> quantum kernel over-spread, advantage unlikely
 d << N -> room for advantage
 - (3) MODEL-COMPLEXITY TEST s_C, s_Q -- now use the ACTUAL labels
 s_C << N and s_Q << N -> both can learn
 s_C ~ N -> classical struggles; check if quantum does better
 - (4) PROJECTED QUANTUM KERNEL
 project states -> local reduced density matrices; keeps g large even as d grows
 this is where P3 sees a real advantage (engineered data, up to 30 qubits)
-

15.5 How WE would test whether our quantum approach helps

OUR ADVANTAGE-TESTING PROTOCOL

1. **Pre-screen with the geometry test.** Compute g_{CQ} between our (optimised) quantum kernel and a suite of tuned classical kernels on the training folds. Report it. If g_{CQ} is small, we predict *no* advantage *before* running the classifier — and then verify.
2. **Report effective dimension d** of the quantum kernel vs. N ; watch for $d \rightarrow N$.
3. **Head-to-head under one protocol.** Same folds, same preprocessing, same tuning budget, same metrics (PR-AUC headline). DeLong / bootstrap CI on every gap.
4. **Scaling mini-study.** PR-AUC vs. $d \in \{4, 6, 8\}$ qubits and vs. training-set size — does the gap grow, shrink, or vanish? ([P10]: only 1/16 studies did this.)
5. **Projected-kernel ablation (advanced).** Try the projected quantum kernel; does it enlarge g_{CQ} and the PR-AUC gap on our task?
6. **State the verdict against the falsification criteria (Part 5).**

SIH JUDGE NOTE

If a judge asks “where is your quantum advantage?”, the correct answer is not a hand-wave about Hilbert space. It is: “We measure it with the [P3] geometry test and a leakage-free head-to-head. On this task we found [X]. We report that result honestly. A genuine asymptotic advantage for arbitrary health data is not known to exist [P10] — our contribution is the method and the data-adaptive map, not a claimed speed-up.”

16 Healthcare-specific issues

16.1 Why early detection is not ordinary classification

- **Asymmetric costs.** Missing an early cancer (false negative) can cost a life; a false positive costs a follow-up test and some anxiety. The decision threshold must reflect that — [P7] recommends lowering τ to hit a chosen *sensitivity floor* (e.g. 0.95).
- **Rare positives.** Screening populations are mostly healthy $\Rightarrow$ heavy imbalance $\Rightarrow$ accuracy and even ROC-AUC mislead; use PR-AUC and recall-at-precision [P5].
- **Weak/early signal.** By definition the disease is not yet obvious; the model must pick up subtle patterns, so calibration and uncertainty matter more than a confident label.
- **Prevalence shift.** The positive rate in your training data $\neq$ the deployment clinic’s; predicted probabilities must be re-calibrated to the deployment prevalence [P7] (TRIPOD+AI).
- **Consequences of action.** A “high risk” output triggers a real intervention (biopsy, referral). Decision-curve analysis quantifies whether acting on the model beats “treat all” / “treat none” [P7].

16.2 The metric set (with the clinical meaning)

Quantity	Meaning	In early detection
Sensitivity / Recall	$P(\text{flag} \mid \text{sick})$	Priority. Set a floor (e.g. ≥ 0.90) and report everything else at that operating point.
Specificity	$P(\text{clear} \mid \text{healthy})$	Drives false-alarm load on the clinic.
False negative	sick, told healthy	The dangerous error — delayed treatment.
False positive	healthy, told sick	Usually recoverable via confirmatory testing [P4][P7].
ROC-AUC	threshold-free ranking quality	Standard, but optimistic under imbalance [P5].

PR-AUC	precision–recall area	<i>Headline</i> metric under imbalance [P5].
Calibration (reliability / ECE / Brier)	do predicted probs match observed frequencies	Clinicians act on the <i>number</i> , not just the rank [P7].
Uncertainty (interval)	how sure is this prediction	Only 1/16 rigorous studies reported it [P10].
Dataset shift / external validation	performance on an independent cohort	Internal AUC $\neq$ deployable [P7][P10].
Decision-curve analysis	net benefit vs. threshold	Translates discrimination into clinical value [P7].
Subgroup / fairness	performance by age/sex/site	Under-reported [P5]; we report at least age & sex strata.

16.3 Privacy, governance, clinical validation

- **Privacy.** Use public de-identified datasets. If real hospital data were ever used: IRB approval, no PII in features, no PII to any external service. [P5] flags privacy as a core constraint; federated / synthetic-data routes are future work, not MVP.
- **Clinical validation.** None of our sources claim it; [P7] is explicit that benchmark-level evidence “does not establish clinical readiness.” Our dashboard must say “**decision support, not diagnosis; not for clinical use.**”
- **Reporting standard.** Follow TRIPOD+AI (external: Collins et al., *BMJ* 2024, 385:e078378) and [P10]’s minimal-requirements table as an appendix / model card.

REMEMBER THIS

Early detection = *rare, weak, high-stakes* positives. Therefore: PR-AUC not accuracy; sensitivity floor not $\tau = 0.5$; calibrated probability + interval not a bare label; external cohort not just internal CV; and a loud “not a diagnosis” disclaimer.

17 Robustness — the experiments

[P7] is the template: it ran ablation, noise, learning curves, feature stability, and internal–external CV, and treated “the model does not beat classical” as a *finding*. [P10] Table 3 lists robustness/sensitivity as “typically untested.” Our robustness battery:

Perturbation	How	What it tells us / paper
Shot / measurement noise	vary shots $M \in \{256, 1024, 4096, \infty\}$	aleatoric floor on the quantum prediction; feeds uncertainty (block 10).
Hardware noise (depolarising)	$p \in \{0, 0.01, 0.05, 0.10\}$ per gate, before & after entanglers	[P7] did this; VQC was robust to $p \leq 0.1$ — but...
Hardware noise (realistic)	add readout confusion + amplitude damping (T_1) + phase damping (T_2) + one coherent over-rotation term	[P7] <i>omitted</i> these and flagged it; we include them and report separately.
Error mitigation	measurement-error mitigation + zero-noise extrapolation on the noisy runs	does mitigation recover PR-AUC? ([P10]: only 1 study distinguished raw vs. mitigated).
Missing data at inference	drop 1–3 features at predict time, impute with train medians	graceful degradation vs. hard failure.
Feature perturbation	add Gaussian noise $\sigma \in \{0.01, 0.05, 0.1\}$ to inputs (both models)	[P7] did the classical half; we do both.
Class-imbalance sweep	train-set positive rate $\{50, 25, 10, 5\}\%$; re-sample <i>in-fold</i> only	Direction B / H3; PR-AUC + recall@precision degradation curves.

Train/test split variation	10 different stratified splits	is the result split-luck? [P5] warns single-split scores are optimistic.
Seed variation	10 optimiser / init seeds for VQC and genetic search	[P2] local minima; report mean $\pm$ SD, not best.
Feature-map variation	ZZ (reps 1–3) vs. alignment-optimised vs. genetic; entanglement pattern $\in \{\text{linear, circular, full}\}$	[P6][P7]; is the winner stable?
Circuit-depth variation	$L \in \{1, 2, 4, 6\}$	[P7] ablation; also a barren-plateau probe.
Embedding-rotation ablation	R_x vs. R_y vs. R_z encoding	<i>mandatory</i> : [P7] shows R_z collapses the VQC to chance.
Internal-external CV	leave-one-stratum-out (e.g. age quartile)	[P7]; generalisation across subgroups.
Feature-selection stability	30 bootstrap replicates, Jaccard index	[P7]: 0.83 = reproducible shortlist; a precondition for interpreting SHAP.
Real-QPU check (advanced)	one short run of the final kernel/circuit on free-tier IBM/IonQ	sim-to-hardware gap, clearly labelled.

17.1 Uncertainty estimation (block 10)

- **Aleatoric (shot noise)**: propagate the binomial standard error of $\langle Z_0 \rangle$ (or of the kernel entries) into the predicted probability.
- **Epistemic (model uncertainty)**: bootstrap ensemble — train $k = 20$ models on resampled training folds (and/or different seeds / re-uploading depths); the spread of their predictions is the epistemic interval.
- **Combine** into one interval; **calibrate** with temperature scaling on a held-out split; **report** reliability diagram + ECE + Brier ([P7]) — now with intervals, which [P10] says almost nobody does.

“WHAT IF OUR MODEL ONLY WORKS ON A PERFECT SIMULATOR?”

That is a likely outcome and it must be *part of the result, not a failure to hide*. Concretely: we report PR-AUC vs. noise level, with and without error mitigation, on both models. If the quantum model’s PR-AUC falls below the classical baseline’s at realistic noise ($p \gtrsim 0.02$ plus T_1/T_2 /readout), we state: *“the quantum model is competitive only in the noiseless / lightly-noised regime; on today’s hardware the classical model is preferable; closing this gap requires error mitigation / better hardware and is future work.”* That is a publishable, honest finding and exactly the kind of statement [P10] says the field is missing.

18 Explainability

18.1 Why it matters

A “high risk” output with no reason is unusable in a clinic and unconvincing to SIH judges. [P5] and [P10] both list explainability as *underexplored* in health QML. [P7] provides the current best-practice recipe.

18.2 Feature importance — global and local

Global

which features drive the model *overall* (mean |SHAP| across the test set).

Local

why *this* patient got *this* score (per-patient SHAP values).

SHAP (SHapley Additive exPlanations)
attributes the prediction to features using a game-theoretic fair-credit rule; `TreeExplainer` is exact for tree models [P7].

18.3 Explaining a quantum model (the hard part)

A VQC/QSVM has no tractable closed form, so [P7]’s approach:

1. Train a **classical surrogate** (gradient-boosted tree) to predict the *quantum model’s own outputs*.
2. Run SHAP on the surrogate.
3. **Audit the surrogate’s fidelity**: label-agreement rate, Spearman & Pearson correlation of probabilities, R^2 . [P7] got 97.1% label agreement, $\rho = 0.82$, $r = 0.91$, $R^2 = 0.27$ — meaning the surrogate tracks the *ranking / decisions* well but not the exact probability scale. So SHAP convergence is read as *global*, not pointwise.

[P7]’s finding: classical-SHAP and quantum-surrogate-SHAP *converged* on the same small feature set (nuclear size / shape descriptors), which matched known cytopathology — evidence the signal is real, not an artefact of one model.

18.4 Limitations of explaining quantum models

- Surrogate can mask local quirks of the true quantum decision boundary [P7].
- Direct circuit-level attribution (gradient/perturbation on the circuit itself) is “active research” [P7] — a possible small extension for our *advanced* version.
- Explanations can themselves be unstable under noise — we test explanation stability (does the top-3 feature set survive the noise sweep?).

18.5 Our explainable dashboard output — designed responsibly

-----+ EARLY-RISK DECISION SUPPORT (NOT A DIAGNOSIS - research MVP) +-----+ Risk band: HIGH Calibrated probability: 0.78 (95% interval 0.64 - 0.88) Operating point: sensitivity floor 0.90 -> threshold 0.31 Top contributing factors (this patient, SHAP): + Fasting glucose strongly increases risk + Systolic BP increases risk + Age slightly increases risk - HDL cholesterol slightly decreases risk Model agreement: classical XGBoost 0.74 quantum QSVM 0.78 Data quality: 0 missing fields in-distribution Guidance: this tool flags patients for EARLIER review by a clinician. It does not diagnose. Confirmed decisions require standard clinical workup. +-----+	
--	--

RESPONSIBLE PRESENTATION RULES

(1) Never show a bare label — always probability + interval + operating point. (2) Always show *why* (top features) and *how sure* (interval). (3) Show classical and quantum outputs side by side — disagreement is information. (4) State data quality (missingness, out-of-distribution flag). (5) Permanent disclaimer: decision support, not diagnosis; not for clinical

use. (6) No patient-identifying data leaves the machine.

19 Quantum optimisation — GEKO and where it belongs

19.1 Why a fixed quantum kernel may not be optimal

The feature map defines the kernel; a mismatched map gives a bad kernel geometry and the QSVM loses ([P7]: ZZ-QSVM AUC 0.727; [P6]: Pauli-ZZ worst on all 9 datasets). Hand-designing a good map is “time-consuming and performance-limited” [P6]. So: *search* for the map.

19.2 Kernel alignment (the cheap optimiser — our MVP)

Kernel-target alignment measures how close a kernel is to the “ideal” label kernel $yy^\top$:

$$\hat{A}(K, yy^\top) = \frac{\langle K, yy^\top \rangle_F}{\|K\|_F \|yy^\top\|_F} \in [-1, 1].$$

Put trainable angles λ inside the feature map, then *gradient-ascend* $\hat{A}$ on the training fold (external: Hubregtsen et al., *Phys. Rev. A* 106:042431, 2022). Cheap: no SVM retrain per step, differentiable, a few dozen iterations.

19.3 GEKO — genetic kernel optimisation (the advanced optimiser, [P6])

Candidate circuits (“individuals”)

gate sequences from $\{X, \sqrt{X}, R_z, \text{CNOT}\}$ (IBM-native, universal) on 5 qubits [P6].

Fitness

supervised = QSVM training accuracy with that kernel; *unsupervised* = max normalised eigenvalue of the kernel matrix (spectral, no labels) [P6].

Selection

keep the fittest individual; update the “base” only if *validation* fitness improves (guards generality).

Mutation

copy the individual k times, mutate each gene with probability $m_p = 0.5$.

Recombination

[P6]’s variant is mainly mutation + selection; classic GAs add crossover.

Iteration (“generations”)

repeat until target fitness or budget; binary-search the gene count for the *smallest* circuit that hits target.

Output

the best kernel circuit.

[P6] results: GEKO kernels match/beat RBF, *consistently* beat Pauli-ZZ, on Moons, XOR, Circles, Wine, Iris, Cancer, Irrigation, **Parkinson’s**, **Drug Classification**; kernel entanglement entropy has *little* correlation with accuracy; supervised $\approx$ unsupervised fitness for these datasets.

```

GEKO loop:
  U0 (initial kernel circuit)
  | mutate x k (m_p = 0.5)
  v
  population {U1..Uk}
  | fitness = supervised: QSVM train acc
  |           unsupervised: max normalised eigenvalue of K
  v

```

```
best Ui  --(val fitness > U0 ?)--> becomes new U0
|
+--- repeat "generations"; binary-search #genes for smallest good circuit
```

19.4 Should GEKO be in our MVP?

MVP vs. ADVANCED — THE OPTIMISER DECISION

MVP: feature-map *selection* = {ZZ reps 1–3, entanglement pattern} grid *plus* **kernel-target alignment**. Reasons: differentiable, fast, one dependency, easy to explain, easy to run leakage-free in nested CV. This already tests H1/H2.

ADVANCED: add **GEKO-style genetic search** as a second optimiser arm. Reasons to keep it advanced-only: (i) it is stochastic and slow (many QSVM re-trains per generation — [P7]’s $O(N^2)$ pain, multiplied); (ii) high overfitting risk on the validation signal — needs an extra held-out confirmation set and a documented compute budget; (iii) it is the part most likely to be seen as “[P6] already did this,” so it must be framed as “GEKO *under a clinical-grade evaluation*.”

Do NOT also bolt on quantum *annealing* optimisation — different hardware, no simulator parity, and [P10] found no advantage for it.

19.5 Kernel alignment $\neq$ accuracy (a caution)

Higher alignment usually helps but not always; and [P6] shows kernel *entropy* is a poor predictor of accuracy. So: optimise alignment, but *select the final map by nested-CV PR-AUC*, not by the alignment number itself.

20 Experiment design

20.1 The three-way comparison

Arm	Model	Role
Classical baseline	LogReg (weak ref) + Optuna-tuned RBF-SVM, XGBoost, LightGBM	the bar to c
Quantum baseline	ZZ-feature-map QSVM + vanilla data-reuploading VQC	“off-the-she
Our method	alignment-optimised QSVM (MVP) + GEKO-optimised QSVM (advanced)	the contribu

20.2 Protocol (identical for every arm)

1. **Split:** patient-level stratified; hold-out test untouched until the end; separate external cohort B.
2. **Cross-validation:** nested — outer 5-fold $\times$ 10 seeds for scoring; inner 3–5 fold for *all* selection (hyper-parameters, *feature map*, threshold, calibration).
3. **Leakage firewall:** scaler, imputer, feature selector, PCA, resampler, feature-map optimiser, calibrator all **fit** on the training fold only.
4. **Equal tuning budget:** same number of trials / wall-clock for classical and quantum tuning (the fairness rule [P7] violated).
5. **Same preprocessing** for classical and quantum arms (same PCA/features).
6. **Metrics:** PR-AUC (headline), ROC-AUC, recall@precision=0.8, Brier, MCC, ECE; 95% CIs (bootstrap + logit for AUC); DeLong test for every quantum-vs-classical gap.

7. **Statistics:** paired tests across folds $\times$ seeds; report effect size + CI, not just p ; Holm correction across the family of comparisons.
8. **Noise experiments** (Part 17) on the winning configuration of each arm.
9. **External validation:** rerun the *whole* protocol from scratch on cohort B.
10. **Advantage screen:** geometry test g_{CQ} + effective dimension d (Part 15).

20.3 Experiment table

Experiment	Purpose	Expected observation	Conclusion it supports
E1 Classical baseline sweep	establish the real bar	tuned RBF-SVM / boosting near ceiling ([P7])	any quantum claim must clear this
E2 ZZ-QSVM vs. baseline	“off-the-shelf quantum” check	ZZ-QSVM at or below baseline ([P6][P7])	fixed maps are not enough
E3 Alignment-QSVM vs. ZZ-QSVM (H1)	does map adaptation help	alignment > ZZ, significant	map adaptation matters
E4 Alignment-QSVM vs. best classical (H2)	the real question	tie or classical slightly ahead (likely)	honest advantage verdict
E5 GEKO-QSVM vs. alignment-QSVM	does heavier search add more	maybe small gain, higher variance, higher cost	cost/benefit of genetic search
E6 VQC vs. QSVM vs. classical	model-family comparison	VQC $\approx$ classical, QSVM depends on map ([P7])	which quantum model to prefer
E7 Imbalance sweep (H3a)	test the “QML needs less balancing” folk claim	likely <i>no</i> robustness advantage ([P3])	confirm/deny a repeated claim
E8 Noise sweep + mitigation (H3b)	hardware realism	PR-AUC degrades with realistic noise; mitigation partially recovers	regime where quantum is viable
E9 Qubit-count scaling ($d \in \{4, 6, 8\}$)	does the gap grow with scale	gap flat or shrinks ([P3])	no evidence of scaling advantage
E10 Seed / split variation	is the result luck	spread ± 0.02 – 0.05 PR-AUC	report distributions not best
E11 Geometry test g_{CQ}	predict advantage before classifying	g_{CQ} small	pre-screen agrees with E4
E12 External cohort B	generalisation	rankings reshuffle, CIs overlap ([P7])	method-level portability, not a fixed model
E13 Explainability convergence	is the signal real	classical-SHAP $\approx$ surrogate-SHAP ([P7])	predictions grounded in real features
E14 Leakage audit	prove the firewall	scores drop vs. a deliberately-leaked run	our numbers are trustworthy

IMPLEMENTATION NOTE

Freeze the protocol *and the falsification criteria* (Part 5) in a pre-registration note committed to the repo *before* running E3–E5. This is the single cheapest credibility move available and directly answers “did you tune until quantum won?”

21 Ablation study

Remove or swap one component at a time; measure the change in nested-CV PR-AUC. This isolates *what each part of the architecture actually contributes*.

#	Configuration	Question it answers
A0	LogReg on raw scaled features	the floor.

A1	Best classical (tuned RBF-SVM / boosting) on scaled features	the real bar ([P7]).
A2	A1 + feature selection	does selection help or hurt the classical bar?
A3	A1 + PCA to d	cost of compressing to the qubit budget (paid by <i>both</i> sides).
A4	QSVM, ZZ map, reps=1, linear entanglement	minimal quantum kernel ([P6] baseline).
A5	A4 + reps $\in \{2, 3\}$ / entanglement $\in \{\text{circular, full}\}$ (grid)	how far does tuning a <i>fixed</i> map get you?
A6	QSVM + kernel-target-alignment optimised map (MVP novelty)	contribution of map adaptation (H1).
A7	A6 + projected quantum kernel	does projection enlarge g_{CQ} / PR-AUC ([P3])?
A8	QSVM + GEKO genetic map (advanced novelty)	does heavier search beat alignment, and at what cost?
A9	VQC, single encoding, no re-uploading, $L=2$	bare variational classifier.
A10	A9 + data re-uploading, $L \in \{2, 3, 4\}$	contribution of re-uploading ([P2]).
A11	A10 + entangling layers vs. none	contribution of entanglement ([P2]).
A12	A10 with R_z encoding instead of R_y	the [P7] embedding trap — expect collapse to chance.
A13	Full proposed pipeline (MVP): A6 + calibration + uncertainty + SHAP	the deliverable.
A14	A13 without calibration	contribution of calibration (Brier, ECE).
A15	A13 without the leakage firewall (deliberately leaked)	size of the leakage inflation — the audit (E14).
A16	A13 with SMOTE outside folds vs. in-fold	the resampling-leakage effect ([P4]).
A17	Stacked hybrid ensemble (classical + quantum probs $\rightarrow$ meta-LR), as in [P7]	does fusing help over the best single model?

21.1 What each component is expected to contribute

- **Feature selection / PCA (A2–A3):** small effect, mostly enabling the qubit budget; must be paid by both sides so it doesn't bias the comparison.
- **Fixed-map tuning (A5–A4):** modest gain; [P6] suggests still below RBF.
- **Map adaptation (A6–A5):** the hypothesised main lever (H1). Could be +0.02–0.08 PR-AUC, or negligible — either is a result.
- **Projected kernel (A7–A6):** [P3] predicts this is where quantum advantage lives, if anywhere.
- **Re-uploading (A10–A9) & entanglement (A11):** [P2] shows both help on structured data; test on health data.
- **R_z encoding (A12):** expected collapse — proves the ablation battery is sensitive and that our design choices are deliberate.
- **Calibration (A14):** negligible discrimination change, large Brier/ECE improvement.
- **Leakage firewall (A15):** A15 should score *higher* than A13 (leakage inflates); the gap is the headline of the leakage audit.
- **Ensemble (A17):** [P7] found the stacked hybrid matched but did not exceed the best single model — expect the same.

KEY IDEA

The ablation is not decoration — it is how we answer “why does each component exist?” to a judge, with a number. Every block in the Part 6 architecture must map to at least one ablation row.

22 MVP vs. advanced version

22.1 MVP — realistically buildable during SIH

A leakage-free hybrid early-detection pipeline on *one* tabular dataset + one external cohort, comparing tuned classical baselines vs. ZZ-QSVM vs. **kernel-target-alignment-optimised QSVM** and a data-reuploading VQC, with PR-AUC + calibration + bootstrap uncertainty + SHAP (classical + surrogate), a noise sweep on a depolarising model, the mandatory ablations (A4–A6, A9–A12, A15), the geometry-test pre-screen, and a read-only risk dashboard. All in simulation.

22.2 Advanced version — research-oriented extensions

GEKO genetic map search (with a compute-budget + overfitting ledger); projected quantum kernel; realistic noise (T_1/T_2 , readout, coherent) + error mitigation; one real-QPU run; a second disease / second dataset pair; subgroup fairness analysis; direct circuit-level attribution; stacked hybrid ensemble; internal–external CV across clinical strata; decision-curve analysis.

22.3 Feature / priority table

Feature	MVP	Advanced	Priority
Dataset A + external cohort B	yes (1 pair)	2nd disease/pair	P0
Leakage firewall + nested CV	yes	yes	P0
Tuned classical baselines (RBF-SVM, XGBoost, LightGBM)	yes	+ CatBoost, RF	P0
ZZ-QSVM baseline	yes	yes	P0
Data-reuploading VQC	yes (L tuned)	+ deeper sweep	P0
Kernel-target-alignment optimised map	yes (core novelty)	yes	P0
GEKO genetic map search	no	yes (+ cost ledger)	P1
Projected quantum kernel	no	yes	P1
PR-AUC + ROC + calibration + Brier + CIs	yes	+ decision-curve analysis	P0
Bootstrap + shot-noise uncertainty interval	yes	+ deep-ensemble	P1
Explainability: classical SHAP + surrogate SHAP + fidelity audit	yes	+ circuit-level attribution	P0/P1
Noise sweep (depolarising)	yes	+ T_1/T_2 /readout/coherent + mitigation	P0/P1
Geometry test g_{CQ} + effective dimension	yes	+ scaling study	P0
Mandatory ablations (A4–A6, A9–A12, A15)	yes	full A0–A17	P0
Real-QPU run	no	yes (1 short run)	P2
Subgroup / fairness analysis	age+sex strata	full IECV across strata	P1
Risk dashboard (read-only)	yes (local web)	+ model card export, PDF report	P0/P1
Pre-registration note in repo	yes	yes	P0
Public code + environment file	yes	+ Docker	P0

DO NOT OVER-BUILD THE MVP

The MVP's value is *rigour*, not feature count. A judge is more impressed by “we ran a leakage audit and here is how much it changed the numbers” than by “we also added a genetic algorithm and a real quantum computer.” Ship the P0 list solidly; add P1 only if P0 is done and verified.

23 Implementation roadmap

Each phase: **build** / **learn** / **output** / **tests** / **failure conditions** / **proceed when**.

23.1 Phase 0 — Literature & gap validation (0.5 week)

Build: a 2-page note confirming [P6][P7] do not already cover our exact combination; pick dataset A + external cohort B; write the pre-registration (protocol + falsification criteria). **Learn:** Parts 1–6, 15, 19 of this document. **Output:** PREREG.md, datasets.md. **Tests:** a supervisor / teammate can restate the gap and the falsification criteria. **Failure:** the exact idea is already published $\Rightarrow$ switch to Direction B (Part 4). **Proceed when:** prereg committed.

23.2 Phase 1 — Dataset & preprocessing (1 week)

Build: loaders, schema validation, patient-level split, the leakage-firewall pipeline (scaler/imputer/selector/PCA as a single sklearn Pipeline fitted per fold). **Learn:** Parts 7, 9, 16. **Output:** data.py, pipeline.py, an EDA notebook (class balance, missingness, feature distributions, post-PCA duplication check). **Tests:** unit test that no transformer sees test data; duplication check passes. **Failure:** dataset has hidden leakage (e.g. a post-diagnosis feature) or B is not truly independent $\Rightarrow$ pick another pair. **Proceed when:** firewall test green.

23.3 Phase 2 — Classical baseline (1 week)

Build: LogReg + Optuna-tuned RBF-SVM, XGBoost, LightGBM inside nested CV; metric + CI + DeLong reporting; calibration. **Learn:** Part 7, 16, 20. **Output:** baseline_results.json, a metrics table with CIs. **Tests:** numbers in the ballpark of published results for the dataset; CV variance sane. **Failure:** baseline is suspiciously perfect (leakage) or terrible (bug). **Proceed when:** baseline is a genuine, tuned bar.

23.4 Phase 3 — Quantum environment (0.5 week)

Build: PennyLane + Qiskit install; statevector + noisy devices; a “hello QSVM” on 2 features; shot-noise sanity check. **Learn:** Part 8, 25. **Output:** quantum.py skeleton, environment file. **Tests:** $K(x, x) = 1$; kernel matrix PSD after regularisation; parameter-shift gradient matches finite-difference. **Failure:** simulator too slow at $d = 8 \Rightarrow$ cap $d = 6$ and/or subsample N . **Proceed when:** sanity checks pass.

23.5 Phase 4 — Quantum feature map (1 week)

Build: angle encoding; ZZ map (reps, entanglement patterns); the overlap circuit; kernel matrix estimation with a chosen shot budget. **Learn:** Parts 9, 10. **Output:** feature_maps.py. **Tests:** ZZ-QSVM reproduces roughly the weak numbers in [P6][P7]; R_z -encoding ablation *does* collapse (proves the harness is sensitive). **Failure:** kernel entries all ≈ 0 (over-spread) $\Rightarrow$ reduce reps / qubits. **Proceed when:** ZZ-QSVM runs end-to-end in nested CV.

23.6 Phase 5 — QSVM + map optimisation (1.5 weeks)

Build: kernel-target alignment optimiser inside the inner CV; select final map by inner PR-AUC; run E2–E4. **Learn:** Parts 11, 19. **Output:** `qsvm.py`, `align_results.json`. **Tests:** alignment increases monotonically-ish; nested-CV score $\leq$ inner-CV score (no leakage); H1 test has a clear verdict. **Failure:** alignment overfits (inner $\gg$ outer) $\Rightarrow$ regularise, fewer trainable angles. **Proceed when:** H1/H2 have honest verdicts with CIs.

23.7 Phase 6 — QNN / VQC (1 week, can parallelise with Phase 5)

Build: data-reuploading VQC, BCE + Adam + parameter-shift, early stop, 10 seeds; run E6. **Learn:** Parts 12, 13, 14. **Output:** `vqc.py`, gradient-variance vs. L plot. **Tests:** training loss decreases; gradient variance not vanishing at $d \leq 8, L \leq 4$; seed spread reported. **Failure:** barren plateau $\Rightarrow$ smaller-angle init, local cost, fewer layers. **Proceed when:** VQC trains reproducibly.

23.8 Phase 7 — Proposed novelty consolidation + GEKO (advanced) (1 week)

Build: full MVP pipeline (A13); *advanced:* GEKO genetic search + compute/overfitting ledger. **Learn:** Part 19. **Output:** `geko.py`, `ledger.md`. **Tests:** GEKO held-out confirmation score consistent with its validation score (else overfit). **Failure:** GEKO too slow $\Rightarrow$ keep it advanced-only, smaller population. **Proceed when:** MVP result table complete.

23.9 Phase 8 — Robustness (1 week)

Build: noise sweep (+ mitigation for advanced), imbalance sweep, seed/split sweeps, ablations A4–A17, external cohort B rerun, geometry test. **Learn:** Parts 15, 17, 21. **Output:** robustness figures, `external_results.json`. **Tests:** curves are smooth; leakage audit (A15) shows the expected inflation. **Failure:** quantum model only works noiseless $\Rightarrow$ *report that* (Part 17 box). **Proceed when:** all planned experiments have data.

23.10 Phase 9 — Explainability (0.5 week)

Build: TreeSHAP on best classical; surrogate + fidelity audit for quantum; feature-stability bootstrap. **Learn:** Part 18. **Output:** SHAP figures, fidelity table. **Tests:** surrogate label-agreement >0.9 ; feature Jaccard >0.7 . **Failure:** surrogate unfaithful $\Rightarrow$ report low fidelity honestly, note as limitation. **Proceed when:** explanations generated + audited.

23.11 Phase 10 — Dashboard (1 week)

Build: FastAPI endpoint that runs the frozen pipeline on an uploaded row; minimal read-only frontend (Part 18 layout); model card. **Learn:** Part 24. **Output:** `app/`. **Tests:** predictions match the offline pipeline exactly; disclaimer always visible; no PII logged. **Failure:** scope creep $\Rightarrow$ ship the static risk card only. **Proceed when:** demo runs offline in <5 s.

23.12 Phase 11 — Final experiments & write-up (1 week)

Build: regenerate every number from a single `make all`; results section; SIH deck (Part 27); judge Q&A rehearsal (Part 28). **Output:** `report.pdf`, `slides`, tagged release. **Tests:** a fresh clone reproduces the headline table. **Failure:** irreproducible $\Rightarrow$ fix seeds / pin versions. **Done when:** reproducible + rehearsed.

IMPLEMENTATION NOTE

Total ≈ 10 – 12 person-weeks for the MVP (P0), $+3$ – 4 for advanced (P1). A 4–6 person team can parallelise Phases 2/3, 5/6, and 9/10. If time is short, cut in this order: real-QPU $\rightarrow$ GEKO $\rightarrow$ projected kernel $\rightarrow$ 2nd dataset $\rightarrow$ decision-curve analysis. *Never* cut the leakage firewall, the tuned baseline, the external cohort, or the falsification criteria.

24 Tech stack

Tool	Status	Why / where	Notes
Python 3.10–3.11	mandatory	everything	match [P7] (3.10) for parity
NumPy	mandatory	arrays, kernel matrices, uncertainty maths	—
Pandas	mandatory	load/clean the health table (block 1–3)	used by [P4][P7]
scikit-learn	mandatory	Pipeline, SVM, LogReg, PCA, feature selection, CV, metrics, calibration	the leakage firewall is a Pipeline per fold
XGBoost / LightGBM	mandatory	strong classical baselines (block C3–C4)	[P7]; tune with Optuna
CatBoost / RandomForest	optional	extra baselines (advanced)	[P7]
Optuna	mandatory	equal-budget hyper-parameter search (classical <i>and</i> quantum)	[P7]; fairness rule
PennyLane	mandatory	VQC, data re-uploading, parameter-shift, <code>lightning.qubit</code> + <code>default.mixed</code>	[P4][P7] used it
Qiskit + qiskit-machine-learning	recommended	<code>FidelityQuantumKernel</code> , <code>ZZFeatureMap</code> , AerSimulator; the QSVM arm	[P7] used it; also gives IBM QPU access
qiskit-aer (noisy)	mandatory	noise sweeps (block 9)	depolarising; advanced: full noise model
DEAP (or a hand-rolled GA)	advanced only	GEKO genetic map search	[P6] approach
SHAP	mandatory	explainability (block 13)	[P7]; TreeExplainer + surrogate
Matplotlib	mandatory	every figure (ROC, PR, reliability, learning curves, gradient variance)	—
Optuna-dashboard / MLflow	optional	experiment tracking	helps reproducibility
FastAPI (or Flask)	mandatory (MVP dashboard)	serve the frozen pipeline; one <code>/predict</code> endpoint	[P4] used Flask-style; keep it read-only
Frontend: plain HTML + htmx / a small React page	mandatory (MVP)	the risk card (Part 18)	no build tooling needed; keep minimal
pytest	mandatory	leakage tests, gradient tests, reproducibility test	—
Docker	optional (advanced)	one-command reproduction	—
Real QPU (IBM Quantum / IonQ free tier)	advanced only	one short run of the final kernel/circuit	clearly labelled; not on the critical path

MINIMAL VIABLE STACK

Python + NumPy + Pandas + scikit-learn + XGBoost/LightGBM + Optuna + PennyLane + qiskit(-aer) + SHAP + Matplotlib + FastAPI + pytest. Everything else is optional or advanced. Pin exact versions in `requirements.txt` ([P7] lists its versions — do the same).

IMPLEMENTATION NOTE

PennyLane vs. Qiskit: use **PennyLane** for the VQC (clean autodiff, parameter-shift) and **Qiskit** for the QSVM kernel (mature `FidelityQuantumKernel`, easy noise models, IBM access). This is exactly the [P7] split. Keep one thin adapter module so the rest of the code doesn't care which backend produced a probability.

25 Code concepts (understand before you implement)

25.1 Programming concepts you must be solid on first

- **scikit-learn Pipeline + Transformer API** — `fit` learns parameters from data, `transform` applies them. The leakage firewall works *because* a Pipeline's `fit` is called only on training folds.
- **Cross-validation objects** — `StratifiedKFold`, `cross_val_predict`, and *nested* CV (a `GridSearchCV`/`OptunaSearchCV` *inside* an outer `cross_validate`).
- **Precomputed kernels** — `SVC(kernel="precomputed")` takes an $N \times N$ matrix; the quantum part only has to *produce* that matrix.
- **Automatic differentiation / parameter-shift** — PennyLane's `qml.grad` gives exact gradients of expectation values via the shift rule; no finite differences.
- **Vectorisation** — build the whole kernel matrix with batched circuit execution, not a Python double loop, or the QSVM is unusably slow.
- **Seeding & determinism** — one global seed threads through splits, resampling, Optuna, and circuit init ([P7]: `seed = 42` everywhere).

25.2 Small examples (skeletons, not production)

(a) Preprocessing as a leak-proof pipeline

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import RobustScaler
from sklearn.feature_selection import SelectKBest, mutual_info_classif
from sklearn.decomposition import PCA

pre = Pipeline([
    ("scale", RobustScaler()),
    ("select", SelectKBest(mutual_info_classif, k=8)),
    ("pca", PCA(n_components=6, random_state=42)),
])
# .fit() ONLY on a training fold, then .transform() others
```

(b) PCA + angle scaling for encoding

```
import numpy as np
Xtr_c = pre.fit_transform(Xtr, ytr)          # fit on train fold
Xte_c = pre.transform(Xte)                  # apply unchanged
lo, hi = Xtr_c.min(0), Xtr_c.max(0)
to_angle = lambda X: np.pi * (np.clip(X, lo, hi) - lo) / (hi - lo)  # -> [0, pi]
Atr, Ate = to_angle(Xtr_c), to_angle(Xte_c)
```

(c) Tuned classical baseline

```
import optuna
from sklearn.svm import SVC
from sklearn.model_selection import cross_val_score, StratifiedKFold
cv = StratifiedKFold(5, shuffle=True, random_state=42)
def obj(trial):
    C = trial.suggest_float("C", 1e-2, 1e3, log=True)
    g = trial.suggest_float("gamma", 1e-4, 1e1, log=True)
    m = SVC(C=C, gamma=g, probability=True, random_state=42)
    return cross_val_score(m, Xtr_c, ytr, cv=cv,
                          scoring="average_precision").mean() # PR-AUC
study = optuna.create_study(direction="maximize")
study.optimize(obj, n_trials=100) # SAME n_trials budget as quantum
```

(d) A quantum feature map + kernel (PennyLane)

```
import pennylane as qml
n = 6
dev = qml.device("lightning.qubit", wires=n)

def feature_map(x, lam=None):
    # lam = trainable alignment angles
    for i in range(n):
        qml.RY(x[i], wires=i) # Ry encoding (NOT Rz - P7 trap)
    for i in range(n):
        qml.CNOT(wires=[i, (i+1) % n]) # circular entanglement
    if lam is not None:
        for i in range(n):
            qml.RZ(lam[i], wires=i)

@qml.qnode(dev)
def overlap(x1, x2, lam):
    feature_map(x1, lam)
    qml.adjoint(feature_map)(x2, lam)
    return qml.probs(wires=range(n))

def kernel_entry(x1, x2, lam):
    return overlap(x1, x2, lam)[0] #  $P(0\dots0) = |\langle \phi_1 | \phi_2 \rangle|^2$ 
```

(e) Kernel-target alignment (our MVP optimiser)

```
import pennylane.numpy as pnp
def kmat(A, lam):
    return pnp.stack([pnp.stack([kernel_entry(a, b, lam) for b in A]) for a in A])
def alignment(lam, A, y):
    K = kmat(A, lam); T = pnp.outer(y, y) # ideal label kernel
    return pnp.sum(K*T) / (pnp.linalg.norm(K) * pnp.linalg.norm(T))
lam = pnp.zeros(n, requires_grad=True)
opt = qml.AdamOptimizer(0.05)
for _ in range(60):
    # ALL on the training fold
    lam = opt.step(lambda l: -alignment(l, Atr, 2*ytr-1), lam)
```

(f) QSVM with the (optimised) precomputed kernel

```
from sklearn.svm import SVC
Ktr = np.array(kmat(Atr, lam)); Kte = np.array(
    [[kernel_entry(a, b, lam) for b in Atr] for a in Ate])
Ktr = 0.5*(Ktr + Ktr.T) + 1e-8*np.eye(len(Ktr)) # symmetrise + jitter -> PSD
qsvm = SVC(kernel="precomputed", C=1.0, probability=True).fit(Ktr, ytr)
p = qsvm.predict_proba(Kte)[: , 1]
```

(g) A data-reuploading VQC

```
@qml.qnode(dev)
def vqc(x, theta):
    for l in range(theta.shape[0]):
        for i in range(n):
            qml.RY(x[i], wires=i)
        for i in range(n):
            qml.Rot(*theta[l, i], wires=i)
        for i in range(n):
            qml.CNOT(wires=[i, (i+1) % n])
    return qml.expval(qml.PauliZ(0))
```

(h) VQC training loop

```
def prob(x, theta): return (vqc(x, theta) + 1) / 2
def bce(theta, X, y):
    p = pnp.stack([prob(xi, theta) for xi in X]); e = 1e-7
    return -pnp.mean(y*pnp.log(p+e) + (1-y)*pnp.log(1-p+e))
theta = 0.1 * pnp.array(np.random.randn(3, n, 3), requires_grad=True) # small init
opt = qml.AdamOptimizer(0.05); best, wait = 1e9, 0
for epoch in range(200):
    theta = opt.step(lambda t: bce(t, Atr, ytr), theta)
    v = bce(theta, Aval, yval)
    if v < best - 1e-4: best, wait = v, 0
    else: wait += 1
    if wait >= 10: break
```

(i) Evaluation + significance

```
from sklearn.metrics import average_precision_score, roc_auc_score, brier_score_loss
def report(y, p):
    return dict(pr_auc=average_precision_score(y, p),
               roc_auc=roc_auc_score(y, p),
               brier=brier_score_loss(y, p))
# DeLong test for the AUC gap between classical p_c and quantum p_q on the SAME y
# (use a small delong.py helper; report gap + 95% CI, not just a p-value)
```

(j) Bootstrap uncertainty interval

```
def bootstrap_interval(fit_predict, Xtr, ytr, x_star, B=200):
    preds = []
    for _ in range(B):
        idx = np.random.choice(len(Xtr), len(Xtr), replace=True)
        preds.append(fit_predict(Xtr[idx], ytr[idx], x_star))
    return np.mean(preds), np.percentile(preds, [2.5, 97.5])
```

IMPLEMENTATION NOTE

Wrap (a)–(f) in a single `fit_fold(train_idx, val_idx, config)` function and call it from the nested-CV driver. Every data-dependent object (`pre`, `lam`, `qsvm`, `theta`, `calibrator`) is created *inside* that function. If any of them is created outside, you have a leak.

26 Research contribution (academic phrasing)

26.1 Problem Statement

Quantum classifiers proposed for early disease detection almost universally employ a fixed, hand-selected quantum feature map. When that map is poorly aligned with the data geometry, the resulting quantum support-vector or variational classifier fails to match even a modestly tuned classical model, and the field’s own systematic reviews report no consistent empirical quantum advantage on clinical tabular data.

26.2 Existing Limitation

Two disjoint bodies of recent work bracket the problem without joining it. (i) Data-adaptive quantum kernel / feature-map optimisation — e.g. genetic search ([P6]) and kernel-target alignment (Hubregtsen et al. 2022) — has been demonstrated only on synthetic and UCI datasets, on a single train/validation/test split family, on ideal (noiseless) simulators, without leakage control for the selection step, without independent-cohort validation, and without any accounting of the search’s compute cost or its propensity to overfit the validation signal. (ii) Clinically-styled QML benchmarking with external validation, calibration, decision-analytic evaluation and explainability ([P7]) has been performed only with a *fixed* ZZ feature map that the authors themselves identify as mis-aligned.

26.3 Research Gap

There is no study that evaluates a *data-adaptive* quantum feature map for an *early-detection* task under a protocol that simultaneously enforces leakage-free nested selection, independent-cohort external validation, imbalance-aware (precision–recall) metrics, realistic-noise robustness testing, multi-seed statistical inference, and a transparent search-cost / overfitting ledger.

26.4 Proposed Method

A reusable hybrid quantum–classical pipeline for early health detection in which the quantum feature map is selected from data by kernel-target alignment (baseline optimiser) or by a hardware-native genetic search (advanced optimiser), with the entire selection procedure nested strictly inside cross-validation folds. The resulting quantum support-vector classifier and a data-re-uploading variational classifier are compared, under one identical protocol and one identical hyper-parameter budget, against Optuna-tuned radial-basis-function support-vector, XGBoost and LightGBM baselines; validated on an independent cohort; stress-tested across class-imbalance ratios, simulated hardware noise levels (with and without error mitigation), qubit counts, and random seeds; pre-screened for prediction advantage using the geometry test of [P3]; and delivered with a calibrated, uncertainty-quantified, SHAP-explained risk output.

26.5 Novelty

Potential novelty (subject to a final literature check at submission time): the specific integration of data-adaptive quantum feature-map optimisation with a leakage-audited, externally-validated, imbalance- and noise-robust evaluation protocol on an early-detection task, together with the first report — to our knowledge — of the optimiser’s compute budget and overfitting behaviour in this setting, and of an empirical gradient-variance / trainability characterisation for the exact circuits used (a measurement [P10] found absent from all sixteen rigorous digital-health QML studies).

26.6 Hypothesis

H1: an adaptive map yields a significantly higher leakage-free PR-AUC than the fixed ZZ map. H2: the adaptive-map model is statistically non-inferior to the best tuned classical model on both the internal test set and the external cohort. H3: the adaptive-map model degrades no faster than the classical baseline under increasing imbalance and increasing noise.

26.7 Experimental Validation

Nested cross-validation (5-fold outer $\times$ 10 seeds, inner selection loop); independent-cohort external validation with the full protocol re-fitted from scratch; DeLong and bootstrap confidence intervals on every quantum–classical gap; the ablation series of Part 21; the robustness battery of Part 17; the advantage screen of Part 15; and a leakage audit (deliberately-leaked run vs. firewalled run) reported as a first-class result.

26.8 Expected Contribution

A public, reproducible pipeline; an honest verdict on H1–H3 for the chosen task, reported whichever way it resolves; a calibrated explainable risk dashboard; and a short decision note — grounded in the [P3] geometry test — on when adapting a quantum feature map is worth the cost for near-term health-data classification.

26.9 Limitations

Simulation-first with at most one confirmatory real-hardware run; small qubit counts that cannot probe the asymptotic regime; public datasets whose “external” cohort is likely of the same research lineage; $O(N^2)$ kernel-estimation cost forcing a capped training-set size; residual overfitting risk in the genetic optimiser, quantified but not eliminated.

POTENTIAL vs. PROVEN CONTRIBUTION

Everything above is a **potential** contribution. It becomes a **proven** contribution only after: (a) a fresh literature search at submission confirms no paper already reports this exact integration on early detection; (b) the experiments are run leakage-free and the results — positive, null, or negative — are reported in full; (c) the code reproduces the headline table from a clean clone. We will not describe any element as “novel” in the SIH pitch without (a), and we will not describe H2 as “confirmed” without (b).

27 SIH presentation — the research story

27.1 The narrative arc (12 beats)

1. **Problem.** Early detection saves lives; the hard cases are the *subtle, early* ones where classical models are least confident and least calibrated.
2. **Existing approaches.** Classical ML (SVM, boosting) is mature but saturated on benchmarks [P7]; QML (QSVM, VQC, hybrid) is proposed as a richer representation [P2][P5].
3. **Current gaps.** Two 2025–26 systematic reviews [P5][P10] and a rigorous 2026 benchmark [P7]: *no consistent quantum advantage*, weak baselines, rampant leakage, almost no external validation, almost no uncertainty, almost no noise testing, and quantum models that use a *fixed* feature map that is often the wrong one [P6][P7].
4. **Our insight.** The quantum model’s weakness is usually the *map*, not the “quantumness.” Adapt the map to the data — but evaluate it to clinical-benchmark standards, which the one paper that adapts maps [P6] does not do.

5. **Proposed architecture.** Leakage-firewalled hybrid pipeline; classical baselines vs. ZZ-QSVM vs. *alignment-optimised* QSVM vs. re-uploading VQC; external cohort; uncertainty; SHAP; noise sweep; dashboard.
6. **Quantum component.** A quantum feature map whose entangling angles are tuned by kernel-target alignment (MVP) or genetic search (advanced), inside nested CV so selection cannot leak.
7. **Innovation.** The *integration*: adaptive map + leakage audit + external validation + imbalance/noise robustness + search-cost ledger + trainability report, on early detection — an empty cell in the literature.
8. **Evaluation.** PR-AUC headline; DeLong CIs on every gap; [P3] geometry test as an advantage pre-screen; pre-registered falsification criteria.
9. **Results.** [fill in]: E3 (does the map help?), E4 (does it match tuned classical?), E8 (does it survive noise?), E14 (how big was the leakage we prevented?).
10. **Healthcare impact.** A pipeline a hospital data team can *inspect* and rerun on their data; a calibrated, explained risk flag for earlier clinician review.
11. **Limitations.** Simulation-first; small scale; public data; decision support, not diagnosis.
12. **Future scope.** Real hardware + mitigation; multi-centre cohorts; biosignals; federated / privacy-preserving training.

27.2 2-minute explanation (say this)

“Early disease detection is about the *subtle* cases. Classical machine-learning models — tuned SVMs and gradient boosting — are already near-perfect on the easy benchmarks, but they can be poorly calibrated and hard to trust on the borderline patients. Quantum machine learning has been proposed as a richer way to represent patient data. But when we read the 2025–2026 systematic reviews, the picture is blunt: there is *no* consistent quantum advantage on health data yet, and most studies have serious method problems — data leakage, weak baselines, no external validation, no uncertainty estimates, and no noise testing. On top of that, almost every quantum classifier uses a *fixed*, hand-picked way of loading data into the quantum circuit — and that fixed choice is often simply the wrong one, which is why the quantum model loses.

Our project does two things. First, instead of a fixed data-loading circuit, we *adapt* it to the data — we tune it to make similar patients map to similar quantum states — and we do this tuning strictly inside cross-validation so it cannot cheat. Second, we evaluate it the way a clinical prediction model *should* be evaluated: against fully-tuned classical baselines, on an independent patient cohort, with precision–recall metrics for the rare positives, with calibrated probabilities and uncertainty intervals, under simulated hardware noise, and with a ‘leakage audit’ that shows exactly how much our safeguards changed the numbers.

The honest goal is not to claim a quantum speed-up — the theory says there isn’t one for arbitrary health data. The goal is a trustworthy, reproducible pipeline plus a real answer to one question: *when you adapt the quantum representation to the data and test it properly, does it close the gap to the best classical model on early detection?* We report that answer whichever way it comes out, and we ship an explainable risk dashboard built on the result.”

27.3 5-minute explanation (structure + extra content)

1. **(0:00–0:45) Hook + problem.** The 2-minute opening, trimmed to the first paragraph. Add one concrete task (e.g. “early type-2 diabetes risk from a routine blood panel”).
2. **(0:45–1:45) What exists & what’s broken.** Show a 4-row table: [P7] (rigorous, fixed map, breast cancer), [P6] (adaptive map, toy data, ideal sim), [P5]/[P10] (reviews: no consistent advantage). Name the five recurring flaws: leakage, weak baselines, no external cohort, no uncertainty, no noise test.
3. **(1:45–2:45) Our idea + architecture.** Walk the Part 6 diagram: highlight the *leakage firewall* box and the *feature-map search* box. State: “the novelty is the combination, not any

single block.”

4. **(2:45–3:45) How the quantum part works.** 30 seconds on the quantum kernel ($K = |\langle \phi(x_i) | \phi(x_j) \rangle|^2$, an overlap you measure); 30 seconds on kernel-target alignment (tune the map so K looks like $yy^\top$); note the [P7] R_z trap as evidence you know the failure modes.
5. **(3:45–4:30) Evaluation & results.** PR-AUC headline; DeLong CIs; geometry-test pre-screen; the ablation that isolates the map’s contribution; the leakage-audit number; the noise curve. Put the H1/H2/H3 verdicts on one slide.
6. **(4:30–5:00) Impact, limits, future.** “Inspectable pipeline + calibrated explained risk flag; simulation-first; decision support not diagnosis; next: real hardware, multi-centre, biosignals.”

SIH JUDGE NOTE

Lead with the *gap and the rigour*, not with “quantum.” The strongest version of this pitch is: “the field’s own reviewers say QML-for-health evaluation is broken; we fixed the evaluation and added one honest technical idea, and here is what we found.”

28 Judge questions — 45 hard ones, answered honestly

28.1 “Why quantum at all?”

Q. Why quantum? What can it do that classical can’t?

Honestly: for arbitrary health data, no *proven* asymptotic advantage exists [P10]. What a quantum feature map *can* do is reach a similarity geometry that no efficient classical kernel reproduces [P3] — but only if the map matches the data. Our contribution is testing whether *adapting* the map delivers a measurable benefit under honest evaluation. We treat “quantum” as a hypothesis to test, not a selling point.

Q. Why not just use XGBoost / LightGBM?

We do — as our headline baseline, fully Optuna-tuned, same budget as the quantum model [P7]. If XGBoost wins, we report that. The point of the project is a fair, leakage-free comparison plus an adaptive quantum map, not a foregone conclusion.

Q. Why not Random Forest / a deep net?

RF is an optional extra baseline. A deep net is overkill for a tabular table of a few hundred patients and would itself overfit [P7]; boosting is the tabular state of the art. For imaging / biosignals a CNN backbone would be the right classical comparator — that’s the advanced version.

Q. Why not a plain SVM?

The RBF-SVM *is* one of our baselines, and it is the classical object our QSVM generalises (same solver, quantum kernel). External work (Kübler et al. 2021) says a bandwidth-tuned RBF often beats quantum kernels — so we tune γ and let it compete.

Q. Where *exactly* is the quantum advantage in your system?

We do not claim one. We *measure* for one: the [P3] geometry test g_{CQ} as a pre-screen, then a leakage-free head-to-head with DeLong confidence intervals. If g_{CQ} is small and the CIs overlap, our stated conclusion is “no advantage on this task” — exactly as [P7] concluded for breast cancer.

28.2 Hardware & scale

Q. Are you actually using a quantum computer?

Development and all headline results are on quantum *simulators* (statevector + noisy), which is standard — [P10] found 131/138 simulation-only health-QML studies used ideal simulators; we go further by using noisy ones. The advanced version includes one short confirmatory run of the final kernel on free-tier real hardware, clearly labelled.

Q. So it’s “just a simulator”?

Yes for now, and we’re explicit about it. A simulator is the correct tool at 4–8 qubits: it is exact, reproducible, and lets us run the noise and ablation battery that real-hardware queue time would make impossible [P7]. The simulation-to-hardware gap is itself one of our reported experiments.

Q. How many qubits, and why that number?

4–8. Reason: angle encoding uses one qubit per input feature, so the qubit count equals the number of PCA components we keep; 4–8 covers $\geq 95\%$ variance for typical clinical tables ([P6] used similar); it keeps circuits shallow enough to train (barren plateaus, [P1]) and simulate fast; and [P7] showed 8 qubits is enough to match classical on WDBC.

Q. Can it scale to more qubits / bigger data?

Not trivially, and we say so. Kernel estimation is $O(N^2)$ circuits [P7]; more qubits can *hurt* via kernel concentration [P3]; trainable circuits free of barren plateaus may be classically simulable anyway (Cerezo et al. 2023). Scale-up is a genuine open research problem, not an engineering detail.

Q. What happens with real hardware noise?

We run a noise sweep (depolarising in the MVP; T_1/T_2 + readout + coherent + crosstalk in the advanced version) with and without error mitigation, and plot PR-AUC vs. noise. [P7] found its VQC tolerated depolarising $p \leq 0.1$ but *omitted* the harder noise types — we include them and report separately. If the model only works noiseless, that is our finding.

28.3 Design choices

Q. Why PCA? Doesn’t it throw away information?

PCA maps an arbitrary feature count onto our fixed qubit budget and decorrelates features [P6][P8]. It is paid by *both* the classical and quantum arms so it doesn’t bias the comparison. We check for the [P10] failure mode — two different-label patients collapsing to the same encoded point — and report it. Feature selection (keep the 8 best raw features) is an alternative arm.

Q. Why this feature map?

The baseline is the ZZ / Pauli map because it is the field standard [P7] and the thing to beat. The *point* of our project is that we don’t commit to it — we adapt the map’s entangling angles by kernel-target alignment, and (advanced) search the gate sequence genetically [P6], all inside nested CV.

Q. Why R_y encoding and not R_z ?

Because [P7] showed an R_z encoding into a Z read-out produces only a global phase — the data becomes invisible and the model collapses to chance (AUC ≈ 0.50). The encoding rotation axis

must not commute with the measurement axis. We include the R_z case as an ablation precisely to demonstrate we understand this.

Q. Why this optimiser (kernel-target alignment / genetic)?

Alignment is differentiable, fast, one dependency, and easy to run leakage-free — ideal for an MVP. Genetic search (GEKO, [P6]) is the advanced arm because it is slower, stochastic, and more prone to overfitting the validation signal — so it needs an extra held-out confirmation set and a compute-budget ledger, which [P6] did not provide.

Q. Why Adam / SPSA for the VQC?

Adam with parameter-shift gradients for simulator work (exact gradients, [P7] used it); SPSA if we move to shots/hardware because it needs far fewer circuit evaluations per step. L -BFGS-B ([P2]) is an option for small training sets.

Q. What is a barren plateau and how do you avoid it?

A region where the loss gradient vanishes exponentially in the qubit count [P1], so training stalls. We avoid it by staying small (≤ 8 qubits) and shallow (≤ 4 layers), using small-angle initialisation, a *local* cost ($\langle Z_0 \rangle$, not a global projector), data re-uploading [P2], and — crucially — we *measure* the gradient variance vs. depth for our exact circuit and report it, which [P10] says no health-QML study has done.

28.4 Evaluation rigour

Q. How do you prevent overfitting?

Nested CV, early stopping on validation loss ([P4]), 10 seeds with reported spread (not best-of), learning curves ([P7]), tiny model capacity, and a held-out confirmation set for the genetic optimiser.

Q. How do you handle class imbalance?

Class weights by default; SMOTE as an arm applied *inside* folds only ([P4] found SMOTE outside folds and naive use cut recall to 76%). Metrics are PR-AUC and recall-at-fixed-precision, not accuracy [P5]. We also run an imbalance sweep to test the folk claim that QML needs less balancing [P2][P4].

Q. How do you avoid data leakage?

A “leakage firewall”: every data-dependent step — scaler, imputer, feature selector, PCA, resampler, *feature-map optimiser*, threshold, calibrator — is fitted only on the current training fold, then applied unchanged ([P7]’s explicit protocol). We also run a *leakage audit*: a deliberately-leaked run vs. the firewalled run, and report the inflation. For context, [P4] removed hard test rows post-hoc; [P10] flags PCA-before-split across the field — we do neither.

Q. How do you validate medically / clinically?

We don’t claim clinical validation — [P7] is explicit that benchmark evidence “does not establish clinical readiness.” We do method-level external validation on an independent cohort (rerun the whole protocol from scratch, [P7]), decision-curve analysis, calibration to deployment prevalence, and TRIPOD+AI-style reporting. Real clinical validation is future work requiring prospective, multi-centre data.

Q. What is sensitivity? Specificity? Why do they matter here?

Sensitivity = $P(\text{flag} \mid \text{sick})$; specificity = $P(\text{clear} \mid \text{healthy})$. In early detection a missed case (low sensitivity) can be fatal while a false alarm is usually recoverable [P4], so we fix a sensitivity floor (e.g. 0.90) and report everything else at that operating point rather than at $\tau = 0.5$.

Q. Your dataset is tiny — how is any of this significant?

It is a real limitation [P5][P7]. We mitigate with nested CV $\times 10$ seeds, bootstrap CIs on every number, DeLong tests on every gap, and an external cohort. Where CIs overlap we say the models are *statistically indistinguishable* rather than picking a winner — exactly [P7]’s stance.

Q. Isn’t your accuracy just the benchmark ceiling / could be noise?

Likely yes on the internal set, which is why PR-AUC (not accuracy) is the headline, why we chose a task where classical baselines are *not* already saturated (unlike WDBC in [P7]), and why every comparison carries a CI and a significance test.

Q. How do you know your explanation is faithful to the quantum model?

We train a classical surrogate on the quantum model’s outputs and *audit* it — label agreement, rank correlation, R^2 ([P7] got 97% / $\rho = 0.82$ / $R^2 = 0.27$, meaning the ranking is faithful but the probability scale less so). We report the audit numbers and read SHAP convergence as *global*, not pointwise.

Q. Why should we believe your baseline is strong?

Because it is Optuna-tuned with a documented trial budget *equal* to the quantum model’s, across RBF-SVM + XGBoost + LightGBM, following [P7]. [P5] and Bowles et al. (2024) identify weak baselines as the #1 way QML papers manufacture fake wins — we pre-empt that.

Q. Did you tune until the quantum model won?

No — and we can prove it: the protocol and the falsification criteria (Part 5) are pre-registered in the repo, committed before the key experiments. Equal tuning budget both sides. If classical wins, that’s in the report.

28.5 Scope, safety, novelty**Q. What is your actual novelty — in one sentence?**

The integration of a *data-adaptive* quantum feature map with a leakage-audited, externally-validated, imbalance- and noise-robust evaluation on an *early-detection* task, plus the first report of the optimiser’s compute cost / overfitting and of an empirical trainability characterisation in this setting.

Q. Hasn’t someone already done this?

Parts of it: [P6] adapts the map (but on toy data, ideal sim, no leakage control, no external cohort, no cost ledger); [P7] does the rigorous evaluation (but with a *fixed*, admittedly mis-aligned map). We are not aware of a paper joining the two on early detection, and we do a fresh literature check at submission. If it exists, we pivot to the imbalance-robustness study (Direction B).

Q. What if classical ML just wins?

Then our deliverables are: the reusable leakage-free pipeline, the honest negative/null result with CIs, the decision note on *why* (geometry test), and the explainable dashboard. [P7] built a well-cited paper on exactly that outcome. A rigorous null is a contribution.

Q. Can your system diagnose patients?

No. It is decision support — it flags patients for *earlier* clinician review. The dashboard says so permanently. It is not for clinical use.

Q. What are the limitations?

Simulation-first; 4–8 qubits (can’t probe the advantage regime); public datasets with a same-lineage “external” cohort; $O(N^2)$ kernel cost forcing capped N ; genetic-search overfitting risk (quantified, not removed); no prospective or multi-centre validation.

Q. Can it run on real hardware today?

The final kernel/circuit — yes, one short run (advanced version). The full experimental battery — not economically (queue time, drift). That’s why simulation is the right development surface.

Q. Why not quantum annealing / D-Wave?

Different hardware, no simulator parity for our comparison, and [P10] found no evidence annealers beat gate-based or classical methods on health data. Out of scope.

Q. What early-detection disease/task, and why?

[choose] a task with (i) a routine, low-cost feature panel, (ii) natural class imbalance, (iii) ≥ 2 public cohorts of the same condition, (iv) classical baselines *not* already at ceiling. Candidates: early type-2 diabetes risk (two cohorts), early chronic kidney disease, PhysioNet early-sepsis, heart-failure onset. We deliberately avoid WDBC because it is saturated [P7].

Q. How is your evaluation different from [P7]?

Same rigour, but (a) an *adaptive* map instead of a fixed ZZ, (b) a *non-saturated* early-detection task with headroom, (c) PR-AUC + imbalance sweep for rare positives, (d) a search compute/overfitting ledger, (e) an empirical trainability report, (f) a leakage-audit experiment.

Q. How is your method different from GEKO [P6]?

GEKO optimises the kernel on toy/UCI data, a single split family, an ideal simulator, with no leakage control for selection, no external cohort, and no report of overfitting or compute cost. We put the selection under nested CV, validate on an independent cohort, add noise and imbalance robustness, and publish the ledger. Also, GEKO is our *advanced* arm; our MVP uses cheaper kernel-target alignment.

Q. What does entanglement actually buy you?

It lets the map/ansatz encode feature *interactions*, and [P2] shows it improves multi-qubit data-efficiency. But [P6] found kernel entanglement entropy is only *weakly* correlated with accuracy — more entanglement is not automatically better. We ablate entanglement pattern (linear / circular / full / none).

Q. How many shots, and how does shot noise affect results?

$M \in \{256, 1024, 4096\}$ in the sweep; $M = 1024$ default. Shot noise adds a $\sim 1/\sqrt{M}$ band to $\langle Z \rangle$ and to kernel entries (which we regularise to stay PSD). This aleatoric uncertainty is propagated into the risk interval (block 10).

Q. What’s your compute budget / carbon story?

Simulator-only MVP runs on a single workstation in ~ 1 – 2 days per full experiment set; we log wall-clock and per-circuit cost ([P7] Table 3 style). The genetic search is the expensive part —

reported explicitly, and off by default.

Q. If the quantum model matches classical but costs 100× more, why bother?

For the MVP, we wouldn't deploy it — and we'd say so. The research value is understanding *whether* map adaptation closes the gap and *under what conditions*, which informs whether the approach is worth pursuing as hardware improves. We present cost honestly alongside accuracy.

Q. Could a quantum-inspired classical algorithm do the same?

Possibly — that's dequantisation [P3]. We include the projected-quantum-kernel ablation and the geometry test partly to see whether our advantage (if any) survives a strong classical kernel. If it doesn't, that's a finding.

Q. What would make you abandon the quantum approach?

If H1 fails (adaptive map doesn't beat ZZ) *and* H2 fails badly on both cohorts *and* the geometry test predicts no advantage — then we'd conclude the quantum arm adds no value for this task and reframe the project around the pipeline and the negative result.

Q. What's the single most important number in your project?

The gap (with CI) between the best classical model's PR-AUC and the adaptive-map quantum model's PR-AUC on the *external* cohort, alongside the leakage-audit inflation number. Those two say whether the result is real and whether quantum helped.

Q. Why should SIH fund / advance this?

Because it produces something the field's own reviewers [P5][P10] say is missing: a reproducible, leakage-audited, externally-validated QML evaluation on early detection, with an adaptive map and honest reporting — useful whether or not quantum wins, and directly reusable by Indian hospital data-science teams on their own datasets.

THE ONE-LINER FOR EVERY TOUGH QUESTION

“We are not claiming a quantum speed-up. We are testing — rigorously and honestly — whether adapting the quantum representation to the data closes the gap to the best classical model on early detection, and we report the answer whichever way it comes out.”

29 Research paper → what I need from it

29.1 [P1] Barren plateaus in QNN training landscapes (McClean et al. 2018)

Year 2018. **Main contribution** for a broad class of random parameterised circuits, gradient variance decays exponentially in qubit count $\Rightarrow$ training stalls. **Study deeply** intro + “gradient concentration” + “practical consequences” + conclusions. **Skim** the Haar-measure proofs and appendices. **Must understand** why expressive/random = untrainable; the depth $O(n^{1/d})$ threshold; local vs. global cost. **Contributes to our prototype** the design rules in Part 14 (small, shallow, small-init, local cost, re-uploading) + the idea of *measuring* gradient variance for our circuit. **Gap it explains** QNN trainability. **Influences architecture?** Yes — bounds our qubit/depth budget.

29.2 [P2] Data re-uploading for a universal quantum classifier (Pérez-Salinas et al. 2020)

Year 2020. **Contribution** re-feeding data between trainable layers makes even a single qubit a universal classifier; entanglement adds data-efficiency. **Study deeply** Sec. 2 (single-qubit structure), Sec. 4 (multi-qubit), cost functions, benchmark tables 1–4. **Skim** the Universal Approximation Theorem proof and BCH algebra. **Must understand** $L(i) = U(\theta_i + w_i \circ x)$; layers $\approx$ hidden neurons; the qubit/re-upload trade-off; weighted-fidelity cost. **Contributes** the design of our VQC arm (Part 13). **Gap** QNN expressivity with few qubits. **Influences architecture?** Yes — our VQC is a re-uploading circuit.

29.3 [P3] Power of data in quantum machine learning (Huang et al. 2021)

Year 2021. **Contribution** classical ML *with data* can match quantum models even on quantum-hard data; the geometry test, dimension test, complexity test; projected quantum kernels; 30-qubit engineered-data advantage. **Study deeply** “Setup + motivating example,” the flowchart (Fig. 1b), geometric difference g_{CQ} , projected quantum kernels, numerical studies. **Skim** the full proofs (Appendices B–L). **Must understand** why accuracy $\neq$ advantage; g_{CQ} small $\Rightarrow$ classical wins; $d \propto N$ collapse; projected kernel idea. **Contributes** our advantage-testing protocol (Part 15) and the projected-kernel ablation (A7). **Gap** lack of a principled advantage test. **Influences architecture?** Yes — adds the geometry-test pre-screen and the projected-kernel arm.

29.4 [P4] Hybrid quantum-classical liver disease detection / QML-Liver (Donaire et al. 2026)

Year 2026. **Contribution** a 2-qubit hybrid (dense layers $\rightarrow$ AngleEmbedding(R_x) + BasicEntanglerLayers) on the Indian Liver dataset; acc 84%, recall 94%, F1 87%; more qubits did *not* help; SMOTE cut recall. **Study deeply** Sec. 4 (preprocessing, model, hyper-parameters), Sec. 5 (results, the 10-qubit and SMOTE ablations). **Skim** the classical ML literature review (Table 1) and optimiser formulas. **Must understand** the hybrid layer layout; why 2 qubits; the class-weight smoothing; **the removal of consistently-misclassified test rows** (a leakage red flag to *not* copy). **Contributes** the MVP hybrid pattern + a cautionary tale on test-set editing and SMOTE. **Gap** missing ablation of the quantum layer’s own contribution. **Influences architecture?** Yes — our MVP resembles it *plus* the missing ablation and firewall.

29.5 [P5] SLR of QML for medical (Setiawan et al. 2026)

Year 2026 (94 studies, 2020–Apr 2025). **Contribution** trends: imaging $>$ signal $>$ tabular data; 87% hybrid; VQC/QNN/QSVM/QCNN dominate; simulation-heavy; weak baselines; imbalance + explainability + external validation underexplored. **Study deeply** Sec. 3.5–3.9 (model categories, algorithms, metrics, discussion) + the future-directions list. **Skim** the dataset catalogue (Table 7) and bibliometrics. **Must understand** the model/architecture distinction; why hybrid dominance is “a compromise not superiority”; the fair benchmarking checklist. **Contributes** the gap table (Part 3) and the framing that this is a methodology problem. **Gap** multiple — it *is* the gap list. **Influences architecture?** Indirectly — justifies the firewall, tuned baselines, PR-AUC, external cohort.

29.6 [P6] Kernel Alignment for QSVM using Genetic Algorithms / GEKO (Creevey et al. 2026)

Year 2026. **Contribution** genetic search over hardware-native gate sequences for QSVM kernels; supervised (accuracy) vs. unsupervised (max eigenvalue) fitness; matches/beats RBF, *consistently* beats Pauli-ZZ on 9 datasets incl. Parkinson’s; kernel entropy $\npropto$ accuracy. **Study deeply** Sec. II (SVM/QSVM background), Sec. III (GEKO method + Algorithm 1), Sec. IV (results, entropy

analysis). **Skim** the full SVM Lagrangian derivation. **Must understand** the GA loop, the two fitness functions, the binary-search on gene count, the native gate set, “update base only if validation improves.” **Contributes** our *advanced* optimiser arm and the direct comparator we must differentiate from. **Gap** fixed feature-map selection (it addresses this) — but leaves leakage control, external validation, noise, cost ledger open. **Influences architecture?** Yes — block Q2 advanced path.

29.7 [P7] Robustness-Oriented Quantum–Classical Hybrid Pipeline for Breast Cancer (Zorlu & Colak 2026)

Year 2026. **Contribution** the current best-practice evaluation: tuned saturating baseline (RBF-SVM + 3 boosting libs), internal–external CV, external cohort (WDBC→WBC), ablation (R_z **embedding collapses the VQC**), depolarising noise, learning curves, feature stability, decision-curve analysis, dual SHAP + surrogate fidelity audit, TRIPOD+AI; VQC *ties* classical, ZZ-QSVM *fails* (AUC 0.727). **Study deeply** *all of Methods* and *all of Results/Discussion* — this is our evaluation template. **Skim** the breast-cancer clinical background. **Must understand** the leakage protocol; equal-tuning-budget rule (which it broke for the QSVM); the embedding-rotation ablation; surrogate SHAP + fidelity audit; “parity, not advantage” framing; the three recommendations (rigorous baselining, full evaluation stack, audited explainability). **Contributes** the entire evaluation design (Parts 16–18, 20–21). **Gap** it uses a *fixed* map — our opening. **Influences architecture?** Heavily.

29.8 [P8] Quantum ML for Genome Disease Diagnosis (Al Khan & Usupova, preprint 2025)

Year 2025, **not peer-reviewed**. **Contribution (claimed)** hybrid quantum-classical framework on UK Biobank / TCGA with amplitude encoding + quantum feature selection; claims higher accuracy + faster computation. **Study deeply** nothing at face value. **Skim** the pipeline outline (preprocessing → amplitude encoding → quantum feature selection → hybrid classifier) as an example of the *genre*. **Must understand** *why it is weak*: hand-waved dataset sizes (“assuming 500,000 participants and 1,000,000 features”), no leakage control described, amplitude-encoding cost ignored, no external validation, preprint disclaimer “data may be preliminary.” **Contributes** a negative example — what a low-rigour QML-health paper looks like, and every trap to avoid. **Gap** it *is* the reproducibility/rigour gap. **Influences architecture?** Only as an anti-pattern.

29.9 [P9] Quantum Machine Learning: A Review and Case Studies (Zeguendry et al. 2023)

Year 2023. **Contribution** a computer-scientist-friendly QML tutorial + 3 case studies: VQC slightly > SVC on Iris (2-class); QSVM ~90% vs. SVM ~85% on WDBC; quanvolutional NN vs. CNN on MNIST. **Study deeply** the encoding section, the SVM→QSVM section, the VQC “layer” definition. **Skim** the broader quantum-algorithms survey and the quanvolutional maths. **Must understand** the pedagogy (use it to teach teammates) *and* that its case studies use *weak baselines / single splits* — a live example of the [P5] critique. **Contributes** teaching material for Parts 8–12; a cautionary example of “QSVM beat SVM by 5%” claims. **Gap** weak baselines / unrealistic benchmarking. **Influences architecture?** No — reference only.

29.10 [P10] Systematic review of QML for digital health (Gupta et al. 2025, npj Digital Medicine)

Year 2025 (4915 screened, 169 eligible, 123 excluded for rigor, 16 synthesised). **Contribution** the definitive current-state verdict: *no consistent performance trend supporting quantum utility*; nearly all models are “linear quantum models”; scaling untested (1/16); uncertainty unreported

(1/16); noise uncharacterised (14/16 compared hardware only to ideal sims); encoding table (angle/amplitude/QRAM costs); the **minimal-requirements framework** (Table 3). **Study deeply** Intro (QML models, encoding Table 1), all Results, all Discussion, Table 3. **Skim** the PRISMA search syntax and per-study appendices. **Must understand** linear vs. non-linear quantum models; “empirical quantum advantage” definition; the four flaws (no scaling, no uncertainty, no noise, ad-hoc metrics); Table 3’s minimum vs. ideal requirements. **Contributes** the backbone of Parts 3, 15–17, 20 and the reporting standard. **Gap** it enumerates them all. **Influences architecture?** Yes — Table 3 is our compliance checklist.

MINIMUM READING PLAN

Everyone reads: [P10] (all), [P7] (Methods + Results), [P5] (Sec. 3.5–3.9 + future directions). **Quantum leads read:** [P2], [P6], [P1] (main text), [P3] (Fig. 1 + geometry section). **ML leads read:** [P4] (Sec. 4–5), [P9] (case studies). **Nobody reads:** every appendix, [P8] beyond skimming.

30 Final cheat sheets

Cheat sheet 1 — Quantum ML terminology

Qubit	unit vector in $\mathbb{C}^2$; n qubits $\rightarrow \mathbb{C}^{2^n}$.
Superposition	non-trivial blend $\alpha 0\rangle + \beta 1\rangle$.
Entanglement	joint state not factorisable into per-qubit states; built by CNOT/CZ.
Gate	unitary (reversible). H , $X/Y/Z$, $R_x/R_y/R_z(\theta)$, CNOT.
Expectation value	$\langle\hat{O}\rangle = \langle\psi \hat{O} \psi\rangle$; estimated from M shots, error $\sim M^{-1/2}$.
Shots	number of circuit repetitions used to estimate a probability / expectation.
Feature map $U(x)$	circuit that encodes classical x into $ \phi(x)\rangle$.
Quantum kernel	$K(x_i, x_j) = \langle\phi(x_i) \phi(x_j)\rangle ^2$.
QSVM	SVM with a quantum kernel; only kernel entries are quantum.
VQC / QNN / PQC	parameterised circuit trained by a classical optimiser.
Ansatz	the trainable part of a PQC (rotation + entangler layers).
Parameter-shift rule	exact gradient: $\frac{1}{2}[\langle O \rangle_{+\pi/2} - \langle O \rangle_{-\pi/2}]$.
Data re-uploading	re-inject x between trainable layers ([P2]).
Barren plateau	gradient variance $\rightarrow 0$ exponentially in qubits ([P1]).
NISQ	noisy, ~ 10 – 100 s qubits, no error correction.
Linear quantum model	data-encoding and trainable parts separate inside the inner product ([P10]); nearly all health QML.
Geometric difference	[P3] pre-screen: small $\Rightarrow$ classical matches/beats quantum.
gCQ	
Projected quantum kernel	kernel from local reduced density matrices; keeps g large as d grows ([P3]).
Dequantisation	a classical algorithm matches the quantum one; advantage gone.
Kernel-target alignment	$\hat{A}(K, yy^\top)$; tune the map to make $K \approx yy^\top$.
GEKO	genetic search for the kernel circuit ([P6]).

Cheat sheet 2 — QSVM

- scale + select/PCA features (fit on train fold) $\rightarrow d$ features (= d qubits)
 - map angles to $[0, \pi]$
 - feature map $U(x)$: Ry encoding (+ optional entangling ZZ / trainable RZ)
 - kernel entry: $|0\dots 0\rangle - [U(x_j)] - [U(x_i)^\dagger] - [\text{measure}] \rightarrow P(0\dots 0) = K(x_i, x_j)$
 - build $N \times N$ K_{train} ($O(N^2)$ circuits!) $\rightarrow$ symmetrise + jitter $\rightarrow$ PSD
 - classical SVC(kernel="precomputed", C) on K_{train}
 - predict: $K(x^*, \text{support vectors}) \rightarrow \text{score} \rightarrow \text{Platt prob}$
 - evaluate INSIDE nested CV; DeLong vs tuned RBF-SVM; report PR-AUC + CI
- FAILURE MODES: Rz encoding (collapse), too many reps/qubits ($K \rightarrow I$),
untuned C / untuned classical comparator, kernel built before split

Cheat sheet 3 — QNN / VQC

```

encode x -> [ Ry(x) ] (re-upload each layer)
ansatz -> L x { Rot(theta) on each qubit ; CNOT ring }      L <= 4, d <= 8
readout -> <Z_0> -> p = (<Z_0>+1)/2
loss -> BCE(p, y)      grad -> parameter-shift      opt -> Adam / SPSA
tricks -> small-angle init (0.1), early stop on val loss, 10 seeds, local cost
report -> learning curve, gradient variance vs L, seed spread (not best-of)

```

Cheat sheet 4 — Quantum feature maps

Map	Form	Notes
Angle	$R_y(x_j)$ per qubit	depth 1, d qubits, our default. Use R_y/R_x , <i>never</i> R_z into a Z read-out.
ZZ / Pauli	angle + $e^{i(\pi-x_i)(\pi-x_j)Z_iZ_j}$, reps r	the field baseline; often mis-aligned ([P7]), worst kernel ([P6]). Tune $r \in \{1, 2, 3\}$ + entanglement pattern.
Alignment-trainable	angle + trainable $R_z(\lambda_i)$ / entangler angles	MVP optimiser; ascend $\hat{A}(K_\lambda, yy^\top)$ on the train fold.
Genetic (GEKO)	evolved sequence of $\{X, \sqrt{X}, R_z, \text{CNOT}\}$	advanced; needs cost ledger + held-out confirmation.
Amplitude	$ \phi(x)\rangle = \sum_k x_k k\rangle$	log d qubits but $O(d)$ depth; ablation only.

Trade-off: more expressive $\Rightarrow$ larger reachable space $\Rightarrow$ (often) smaller useful g_{CQ} and worse trainability ([P1][P3]). Select the final map by *nested-CV PR-AUC*, not by alignment or entropy.

Cheat sheet 5 — Quantum kernels

```

K(x_i, x_j) = |<phi(x_i)|phi(x_j)>|^2 in [0,1], K(x,x)=1
estimate: run overlap circuit M times, count all-zeros / M
cost:      O(N^2) circuit families x M shots
pitfalls:  - shot noise -> K not PSD (fix: 0.5(K+K^T) + eps*I)
           - too expressive / too many qubits -> K -> I, generalisation dies (P3)
           - hardware noise -> K flattens
tests:     geometry difference g_CQ vs tuned classical kernels (P3 pre-screen)
           effective dimension d vs N (d ~ N => no advantage)
classical rival: tuned RBF often beats quantum kernels (Kubler 2021) -> TUNE gamma

```

Cheat sheet 6 — Healthcare metrics

Sensitivity / recall	TP/(TP+FN)	priority in screening; set a floor, report at that point.
Specificity	TN/(TN+FP)	false-alarm load.
PR-AUC	area under precision-recall	headline under imbalance ([P5]).
ROC-AUC	rank quality	standard but optimistic under imbalance.
Brier / ECE / reliability	probability calibration	clinicians act on the number ([P7]).
Decision-curve analysis	net benefit vs. threshold	is acting on the model worth it ([P7]).
Uncertainty interval	epistemic + shot noise	almost never reported ([P10]).
External-cohort metric	same battery on cohort B	internal $\neq$ deployable ([P7][P10]).

DeLong / bootstrap significance of a gap don't over-read 0.5% ([P7]).
CI

Never headline *accuracy*. Never use $\tau = 0.5$ blindly for a rare disease.

Cheat sheet 7 — Research gaps (status)

Gap	Status	Note
Consistent quantum advantage on health data	B	[P10]: none found; reframe around method.
Weak classical baselines	B	tune RBF-SVM + boosting, equal budget ([P7]).
Ideal-simulator dependence	C	add realistic noise + mitigation.
Barren plateaus (measure)	C	report gradient variance for our circuit — nobody does.
Barren plateaus (solve)	A	use known design rules.
Small datasets / class imbalance	C	PR-AUC, in-fold resampling, imbalance sweep.
Data leakage	B	leakage firewall + leakage audit as a deliverable.
External validation	B	method-level external cohort ([P7]).
Uncertainty estimation	C	bootstrap ensemble + shot noise - $\hat{\epsilon}$ interval.
Fixed feature-map selection	C	our #1 lever — adaptive map + honest eval.
Reproducibility	B	prereg + pinned versions + one-command re-build.
Standardised evaluation	B	adopt [P10] Table 3 + TRIPOD+AI.
Circuit depth / hardware limits	A	accept the constraint (4-8 qubits).

A = solved, B = partial/crowded, C = realistically open, D = too hard for SIH.

Cheat sheet 8 — Our proposed architecture

```
data A + data B(external)
-> validate -> patient-level split
-> [ NESTED CV : leakage firewall ]
  clean/impute -> scale -> select -> PCA(d) -> imbalance(in-fold)
  CLASSICAL: LogReg | RBF-SVM* | XGBoost* | LightGBM*      (*Optuna, equal budget)
  QUANTUM:   encode(Ry) -> map search { ZZ | alignment(MVP) | GEKO(adv) }
             -> Q3a QSVM (quantum kernel -> SVC) | Q3b re-uploading VQC
-> comparative metrics (PR-AUC headline, DeLong CIs)
-> robustness (noise / seed / imbalance sweeps, ablations, Rz trap)
-> uncertainty (bootstrap + shot noise) -> calibration (in-fold)
-> external cohort B (full protocol re-fit)
-> explainability (classical SHAP + surrogate SHAP + fidelity audit)
-> risk output (band + calib prob + interval + top features + model card)
-> dashboard (read-only: "decision support, not diagnosis")
NOVELTY = adaptive map + leakage audit + external validation
          + imbalance/noise robustness + search-cost/overfitting ledger
          + trainability report , all on an EARLY-DETECTION task
```

Cheat sheet 9 — Implementation roadmap

P0	literature + prereg (falsification criteria!)	0.5 wk
P1	data + leakage-firewall pipeline + EDA	1 wk
P2	tuned classical baseline (Optuna, nested CV, CIs)	1 wk
P3	quantum env + sanity checks (K(x,x)=1, grad check)	0.5 wk
P4	feature map (ZZ) + kernel estimation + R _z -trap ablation	1 wk
P5	QSVM + kernel-target alignment + E2-E4	1.5 wk
P6	data-reuploading VQC + gradient-variance report	1 wk (parallel w/ P5)
P7	MVP consolidation (+ GEKO advanced + ledger)	1 wk

P8 robustness: noise/imbalance/seed sweeps, ablations, B 1 wk
P9 explainability (SHAP + surrogate + fidelity audit) 0.5 wk
P10 dashboard (FastAPI + read-only card) 1 wk
P11 final experiments + report + deck + judge Q&A drill 1 wk
NEVER CUT: leakage firewall, tuned baseline, external cohort, falsification criteria
CUT ORDER IF LATE: real-QPU -> GEKO -> projected kernel -> 2nd dataset -> DCA

Cheat sheet 10 — SIH judge answers (rapid-fire)

Why quantum?	Testing whether an adaptive quantum representation closes the gap to classical; not claiming a speed-up.
Why not XGBoost?	It's our headline baseline, equally tuned. If it wins, we report it.
Quantum advantage where?	Measured via [P3] geometry test + DeLong CIs; likely "none on this task," reported honestly.
Real quantum computer?	Simulators (incl. noisy) for all headline results; 1 short real-QPU run in the advanced version.
How many qubits / why?	4–8 = PCA components = angle-encoding qubits; shallow enough to train ([P1]) and simulate.
Noise?	Noise sweep $\pm$ mitigation; if it only works noiseless, that's the finding.
Why PCA?	Maps features onto the qubit budget; paid by both arms; post-PCA duplication checked.
Why this feature map?	We <i>don't fix</i> it — alignment (MVP) / genetic (adv) search inside nested CV.
Why R_y not R_z ?	R_z into a Z read-out is invisible — [P7] showed AUC collapses to 0.5.
Barren plateau?	Vanishing gradients in qubit count ([P1]); avoided by small/shallow/local-cost; we measure gradient variance.
Overfitting?	Nested CV, early stop, 10 seeds (report spread), tiny capacity, held-out confirmation for GEKO.
Imbalance?	Class weights + in-fold SMOTE arm; PR-AUC not accuracy; imbalance sweep.
Leakage?	Firewall (fit-on-train-fold-only) + a leakage-audit experiment reporting the inflation.
Medical validation?	Method-level external cohort + DCA + calibration + TRIPOD+AI; <i>not</i> clinical validation — future work.
Sensitivity/specificity?	Set a sensitivity floor; report at that operating point, not $\tau = 0.5$.
Explainable?	Classical SHAP reference + surrogate SHAP for quantum + fidelity audit ([P7]).
Actual novelty?	Adaptive map + clinical-grade leakage-free eval + cost/overfitting ledger + trainability report, on early detection.
Already done?	[P6] adapts (toy, ideal); [P7] evaluates (fixed map). Not joined on early detection — rechecked at submission.
If classical wins?	Deliverables become the pipeline + honest null + decision note + dashboard ([P7] did exactly this).
Diagnose patients?	No — decision support only; permanent disclaimer.
Limitations?	Simulation-first; small scale; public data; $O(N^2)$ kernel cost; GEKO overfitting risk.
Scale / real hardware?	Final circuit yes (1 run); full battery no; scale-up trainability is an open problem.

End of document. Optimise for understanding $\rightarrow$ research gap $\rightarrow$ novelty $\rightarrow$ architecture $\rightarrow$ implementation $\rightarrow$ experimentation $\rightarrow$ SIH defence.